US012397592B2

(12) **United States Patent**
**Sukegawa et al.**

(10) **Patent No.: US 12,397,592 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **TIRE**

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( * ) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/706,844**

(22) PCT Filed: **Jun. 22, 2022**

(86) PCT No.: **PCT/JP2022/024946**
§ 371 (c)(1),
(2) Date: **May 2, 2024**

(87) PCT Pub. No.: **WO2023/105824**
PCT Pub. Date: **Jun. 15, 2023**

(65) **Prior Publication Data**

US 2025/0010667 A1 Jan. 9, 2025

(30) **Foreign Application Priority Data**

Dec. 9, 2021 (JP) ................................ 2021-200133

(51) **Int. Cl.**
***B60C 19/00*** (2006.01)
***B60C 13/00*** (2006.01)

(52) **U.S. Cl.**
CPC ........... ***B60C 19/00*** (2013.01); ***B60C 13/001*** (2013.01)

(58) **Field of Classification Search**
CPC . B60C 13/001; B60C 19/00; B60C 2019/004; B60C 23/0493
See application file for complete search history.

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(57) **ABSTRACT**

A tire according to the present disclosure includes a tire body and a communication device embedded within the tire body or mounted on an inner surface of the tire body. The tire includes a label area that is located on an outer surface of the tire body and indicates the position of the communication device. The label area includes a light emission area.

**8 Claims, 16 Drawing Sheets**

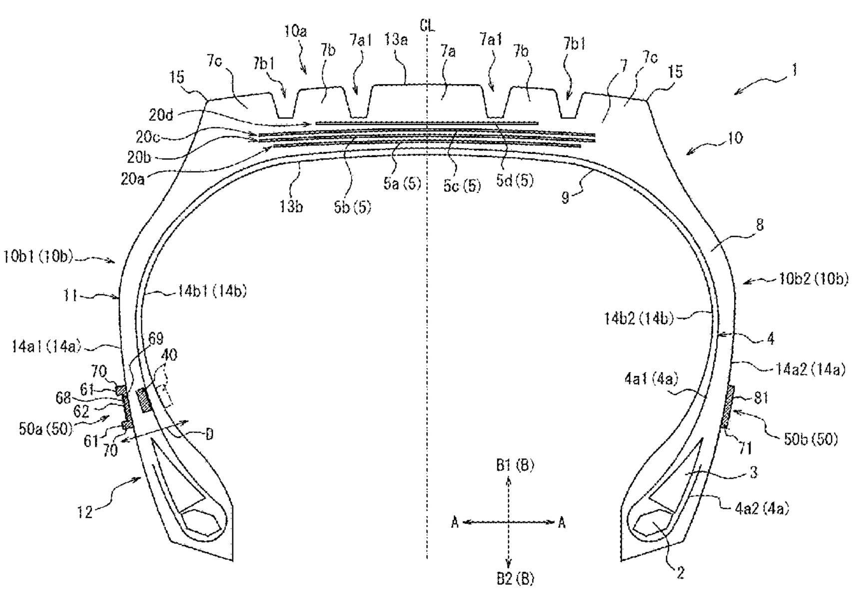

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FIG. 1

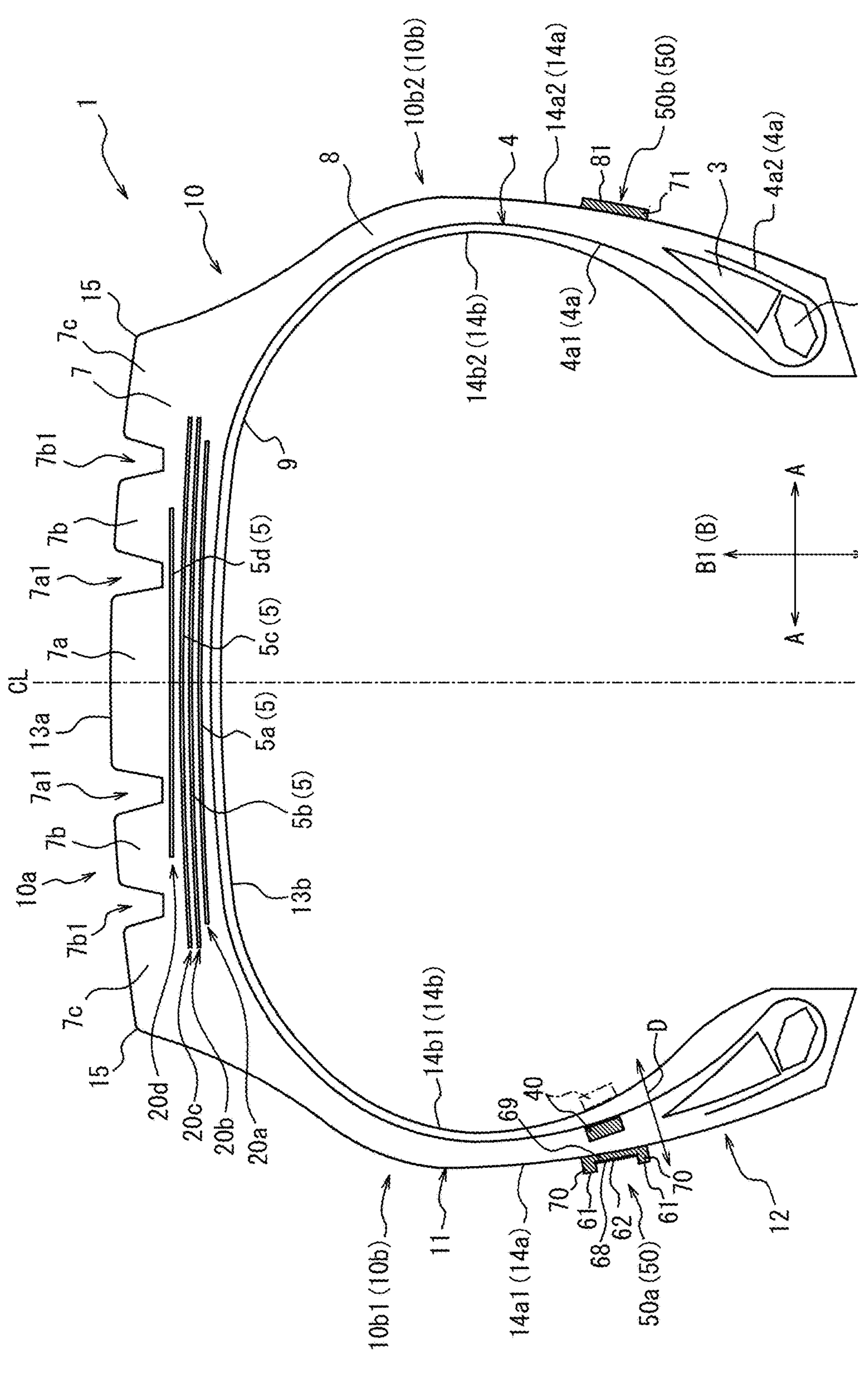

1
10
10b2 (10b)
14a2 (14a)
50b (50)
81
71
4
3
4a2 (4a)
8
2
15
7c
7
14b2 (14b)
4a1 (4a)
9
7b1
7b
7a1
7a
CL
13a
5d (5)
5c (5)
5a (5)
5b (5)
13b
10a
20d
20c
20b
20a
14b1 (14b)
D
40
69
70
61
62
68
50a (50)
14a1 (14a)
11
10b1 (10b)
12
A
B1 (B)
B2 (B)

FIG. 2

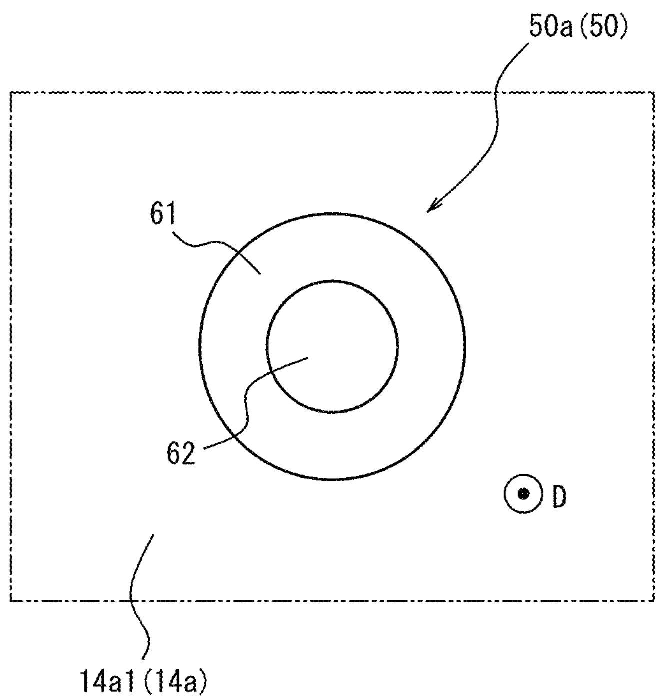


FIG. 3

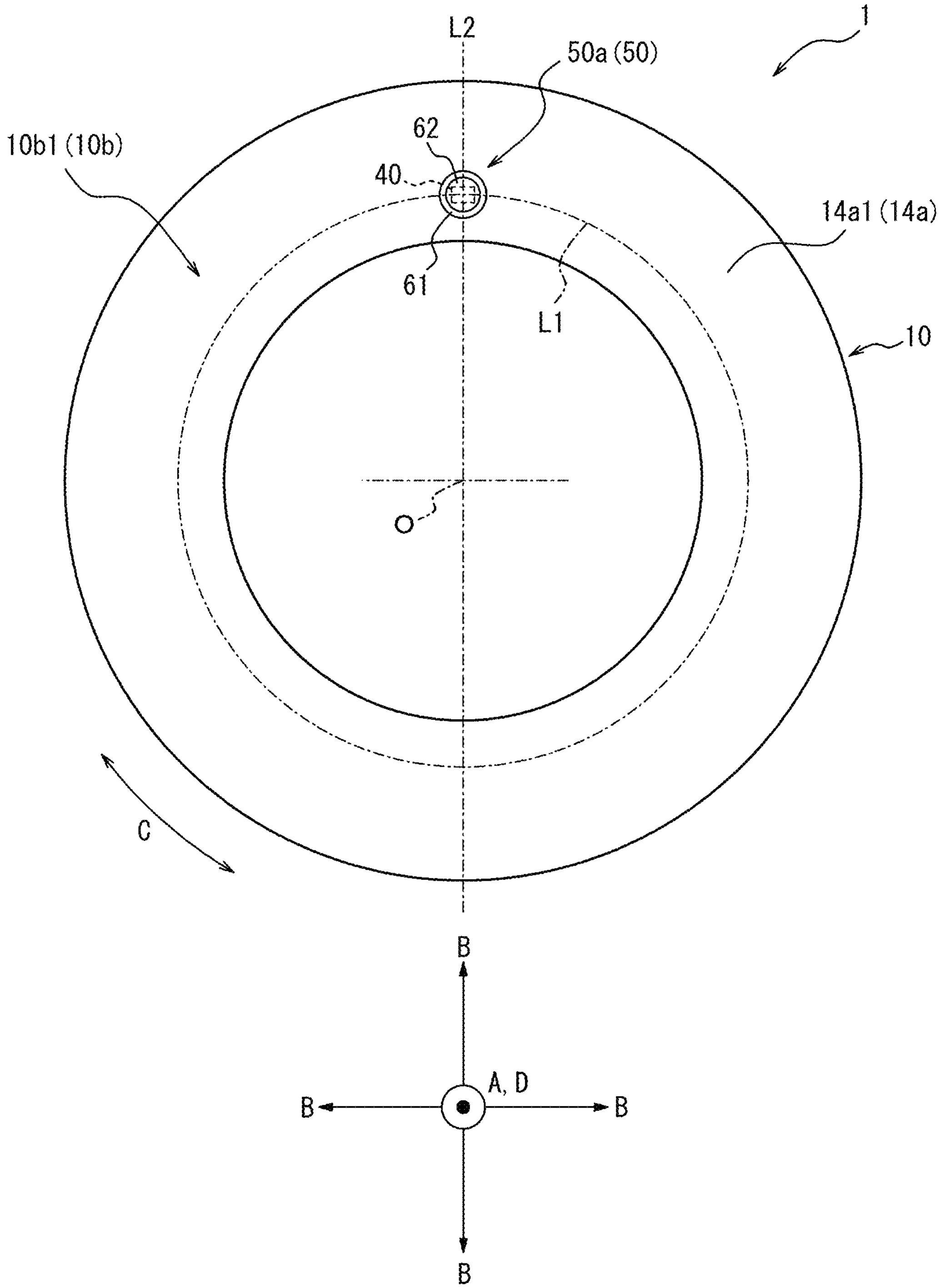


FIG. 4A

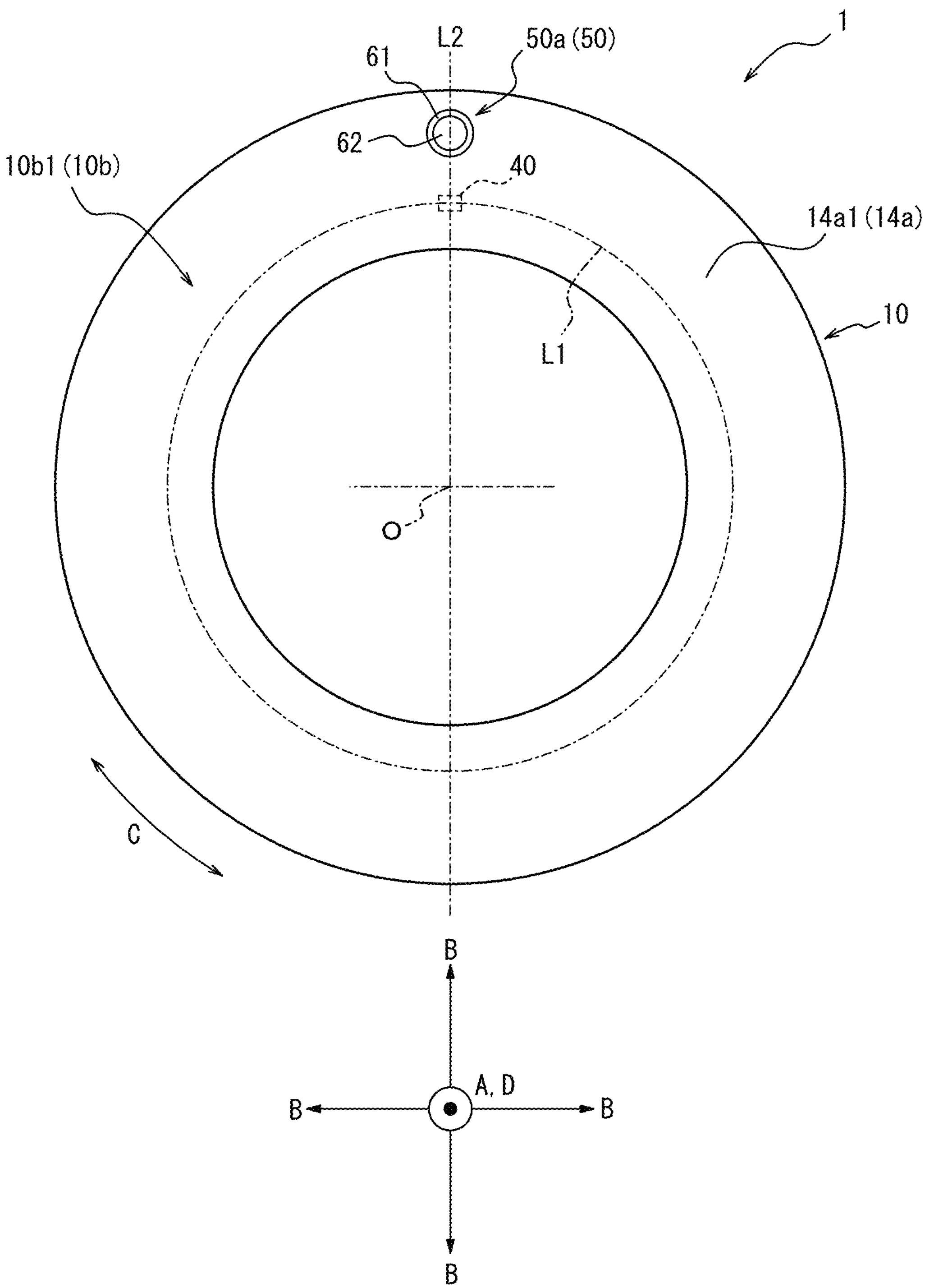


FIG. 4B

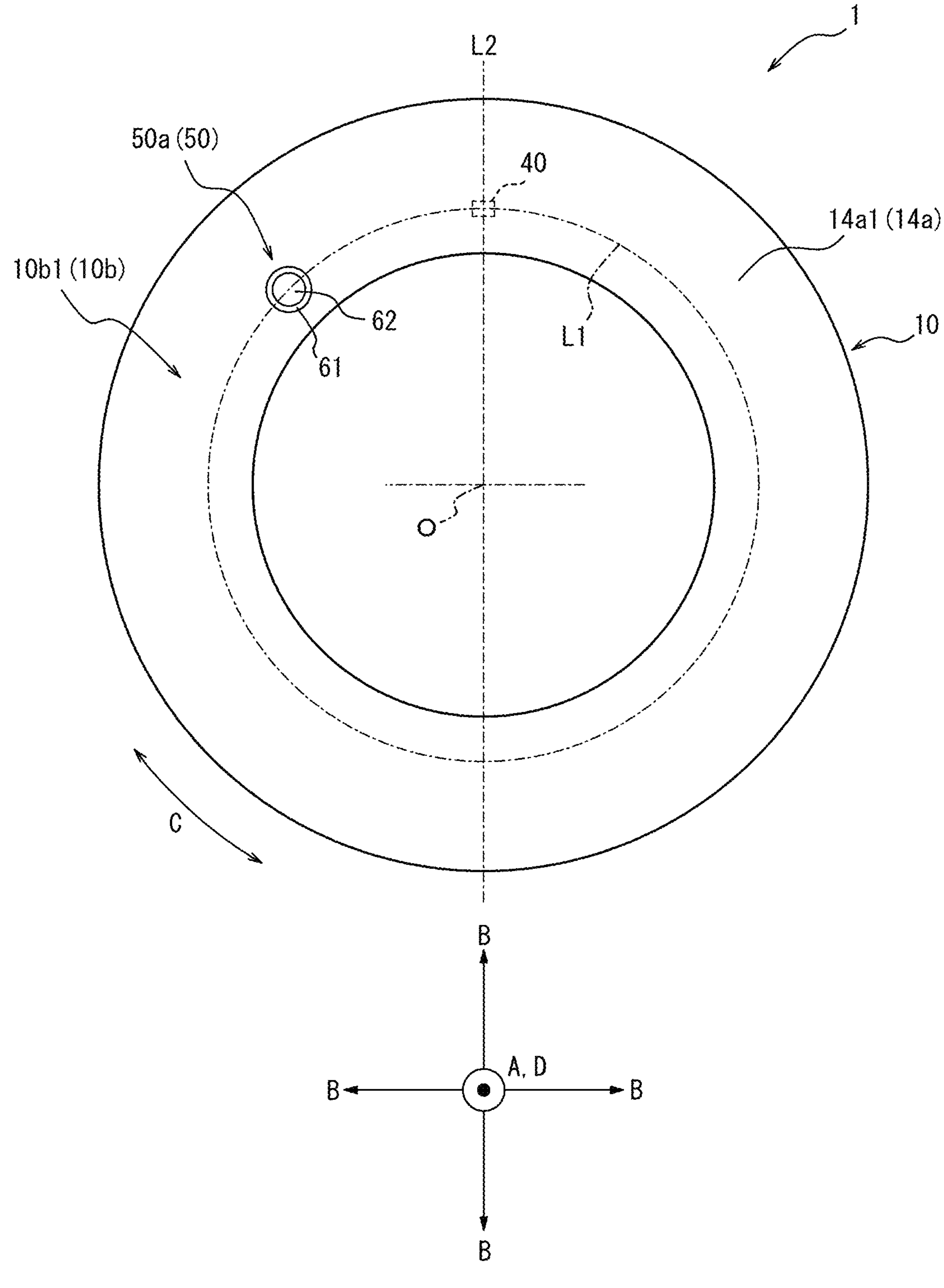


FIG. 4C

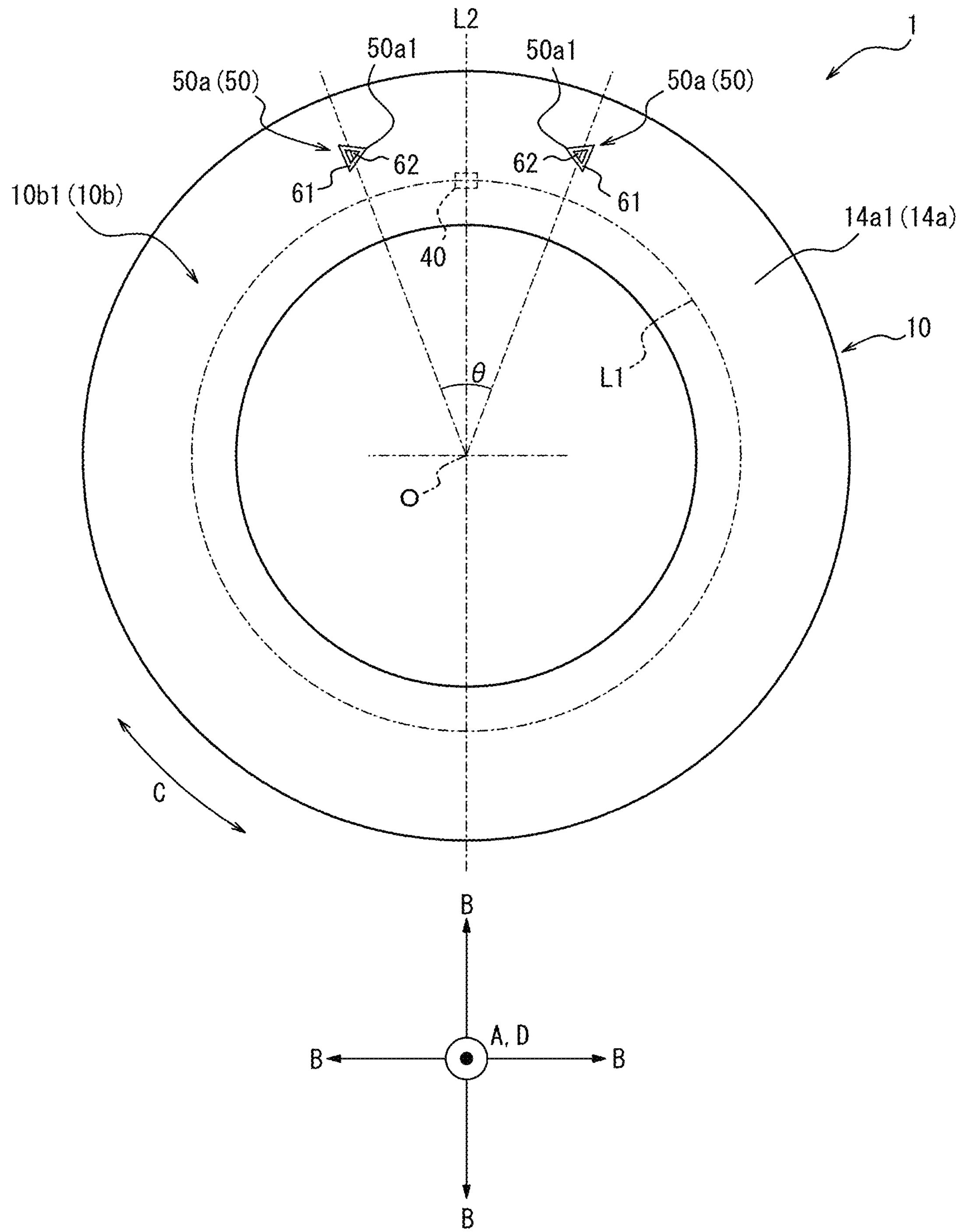

L2
50a1
50a1
1
50a (50)
50a (50)
62
62
61
61
10b1 (10b)
40
14a1 (14a)
10
θ
L1
O
C
B
A, D
B
B
B

FIG. 4D

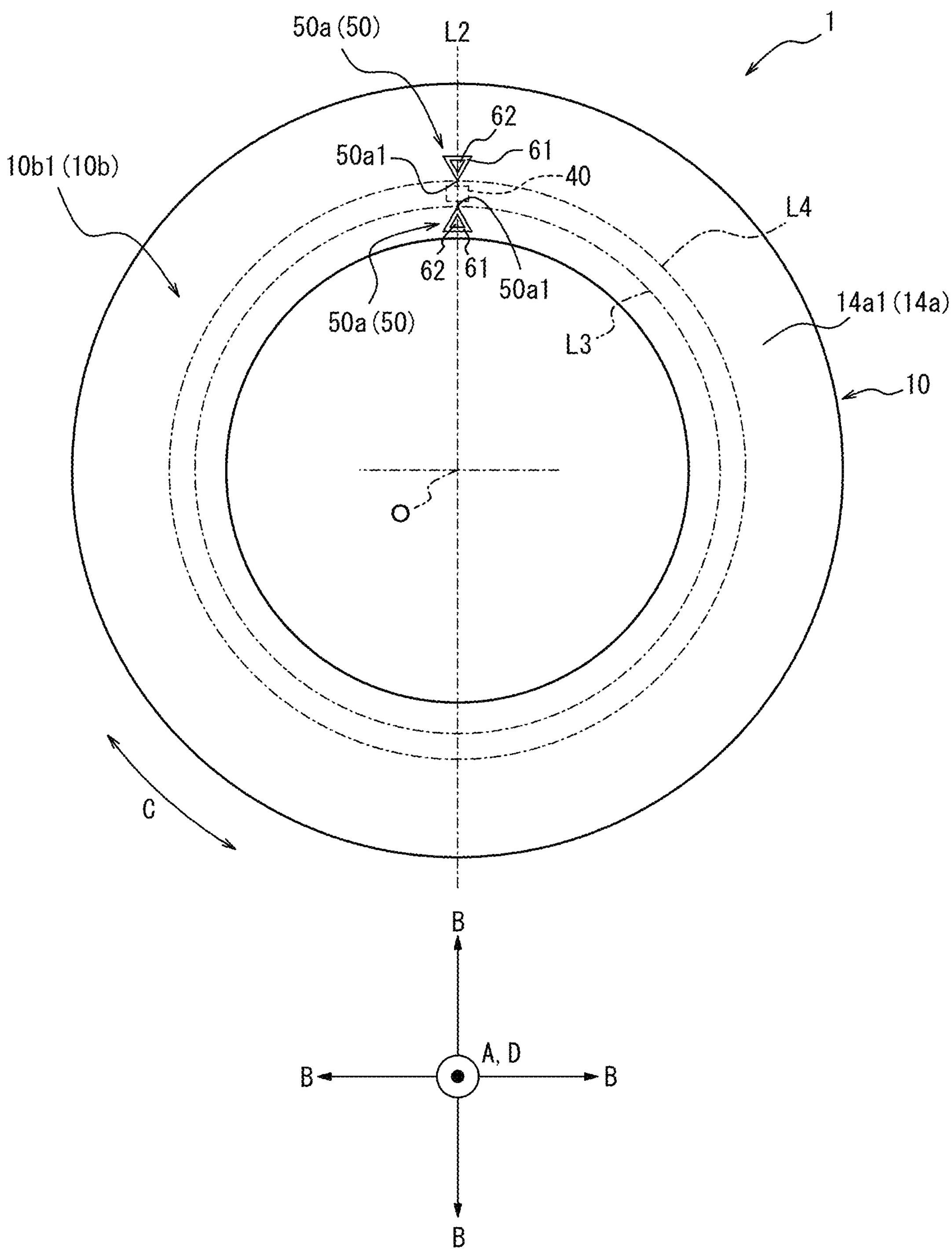

1
50a (50)
L2
62
61
50a1
40
10b1 (10b)
L4
62
61
50a1
50a (50)
L3
14a1 (14a)
10
O
C
B
A, D
B
B
B

FIG. 4E

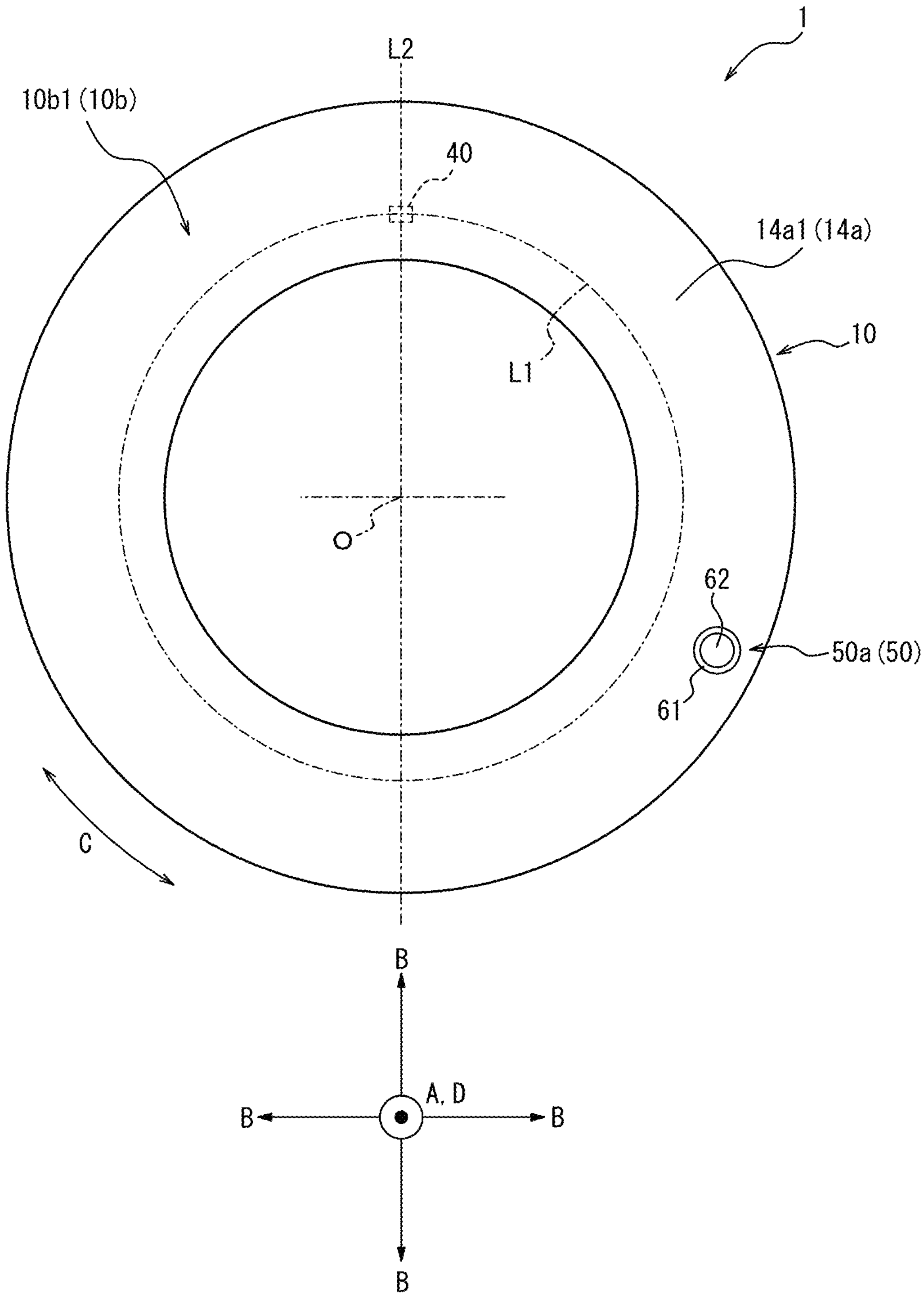


FIG. 5

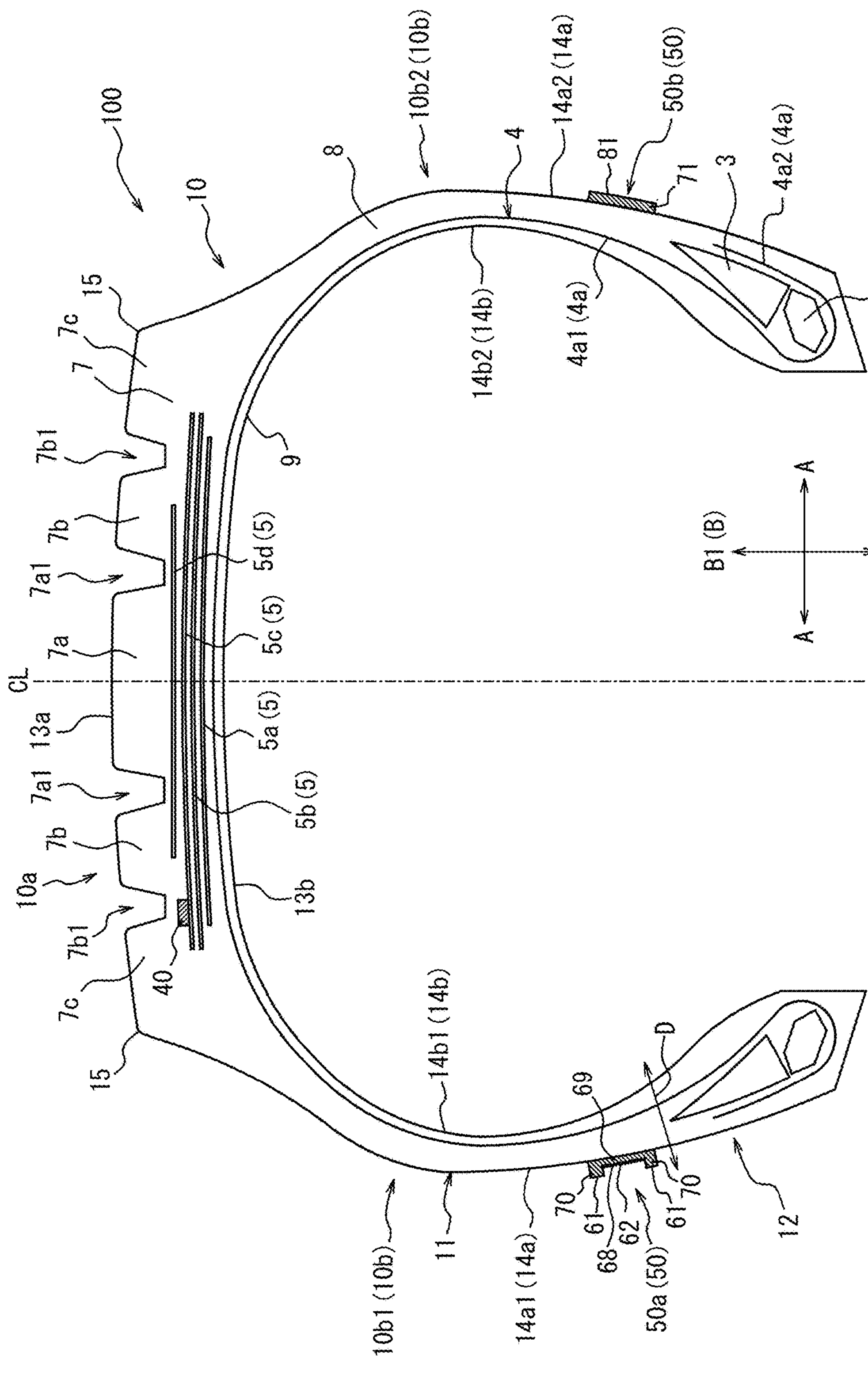
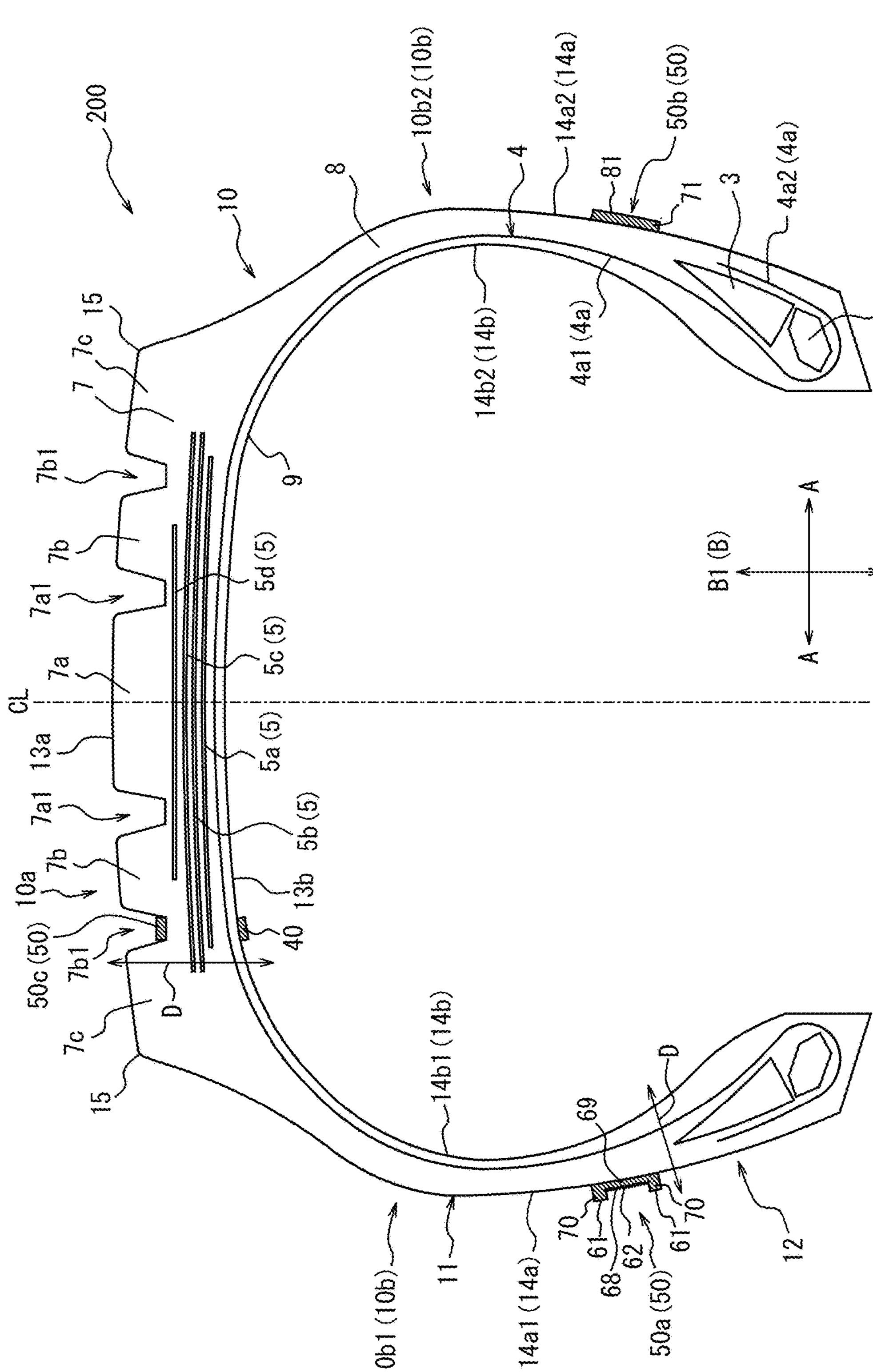

100
10
10b2 (10b)
14a2 (14a)
50b (50)
81
71
8
4
3
4a2 (4a)
2
15
7c
7
14b2 (14b)
4a1 (4a)
7b1
9
7b
5d (5)
7a1
7a
5c (5)
CL
13a
5a (5)
5b (5)
13b
10a
40
14b1 (14b)
D
69
10b1 (10b)
11
14a1 (14a)
70
61
68
62
50a (50)
12
A
B1 (B)
B2 (B)

FIG. 6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10
10b2 (10b)
14a2 (14a)
50b (50)
8
4
81
71
3
4a2 (4a)
2
14b2 (14b)
4a1 (4a)
15
7c
7
9
7b1
7b
5d (5)
7a1
5c (5)
7a
CL
13a
5a (5)
5b (5)
13b
10a
50c (50)
7b1
40
D
7c
15
14b1 (14b)
D
69
11
10b1 (10b)
14a1 (14a)
70
61
68
62
50a (50)
61
70
12
B1 (B)
B2 (B)
A
A

FIG. 7

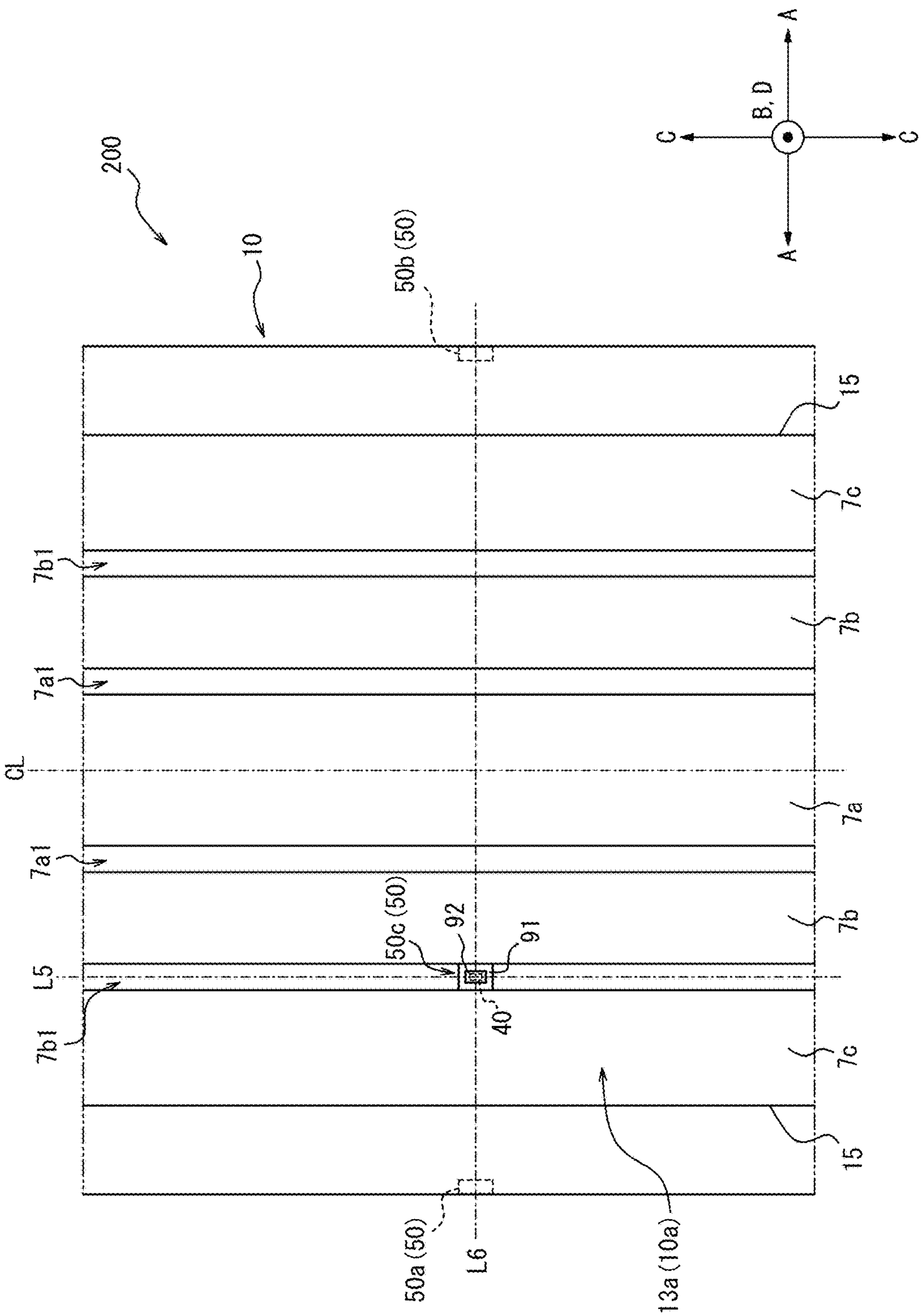
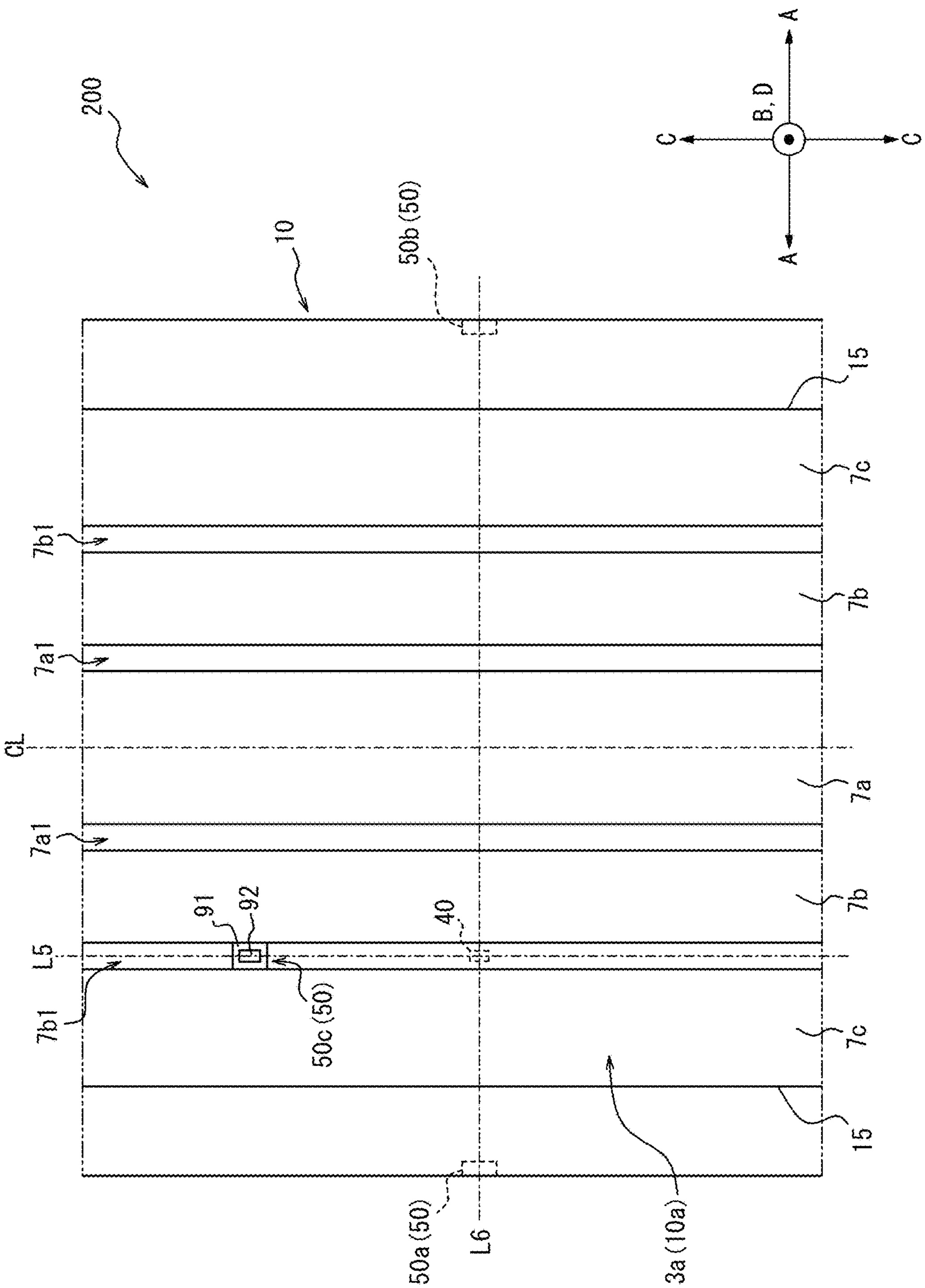

200
10
50b (50)
A
B, D
C
C
A
15
7c
7b1
7b
7a1
CL
7a
7a1
7b
50c (50)
92
91
L5
40
7b1
7c
15
50a (50)
L6
13a (10a)

FIG. 8A


200
10
50b (50)
A
B, D
C
C
A
15
7c
7b1
7b
7a1
CL
7a
7a1
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L5
7b1
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7c
15
50a (50)
L6
13a (10a)

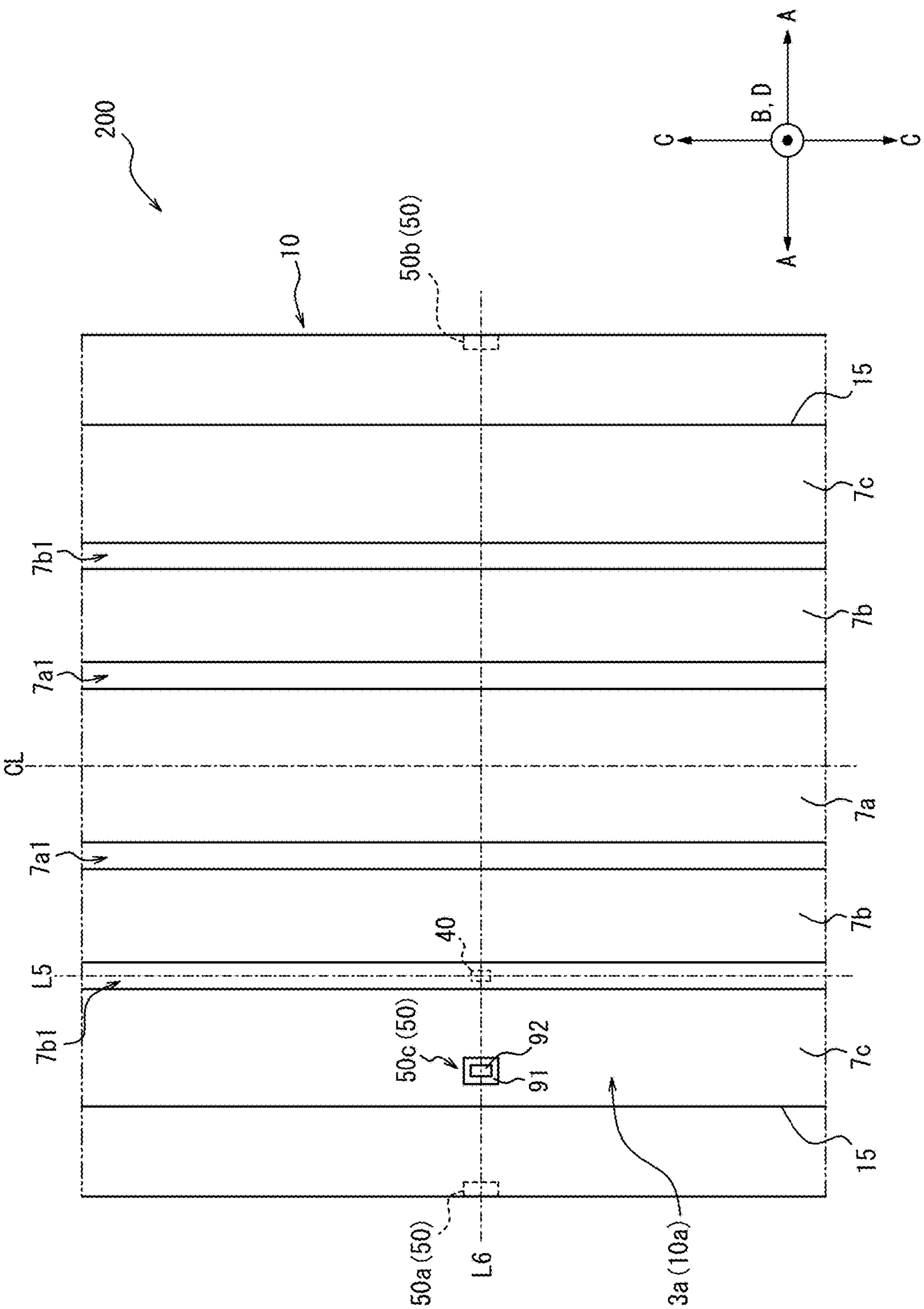
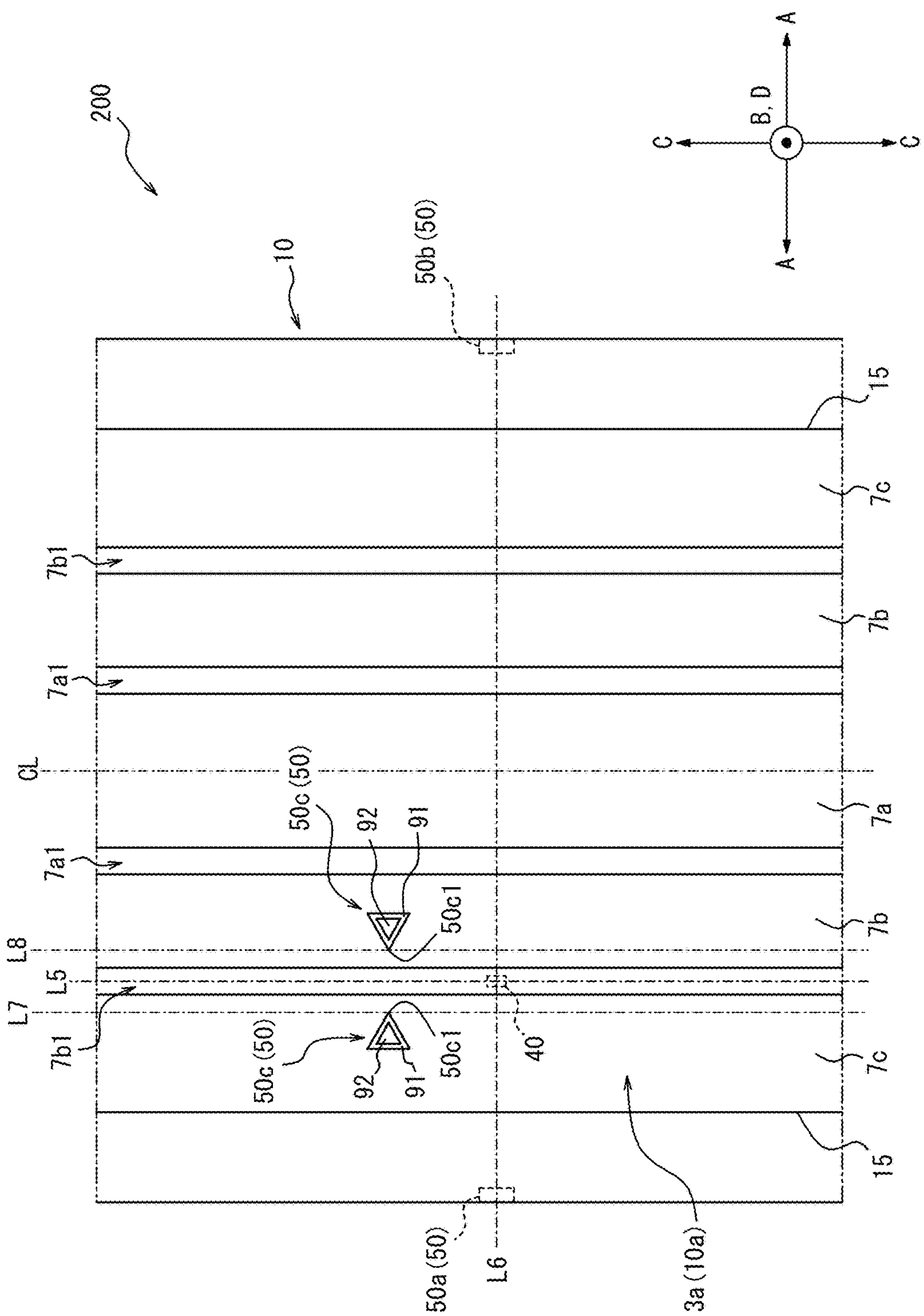

200
10
50b (50)
15
7c
7b1
7b
7a1
CL
7a
7a1
7b
40
L5
7b1
50c (50)
92
91
7c
15
50a (50)
L6
13a (10a)
A
B, D
C
C
A

FIG. 8B

FIG. 8C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A
B, D
C
15
7c
7b1
7b
7a1
CL
50c (50)
92
91
7a
50c1
L8
L5
L7
40
50a (50)
L6
13a (10a)

FIG. 8D

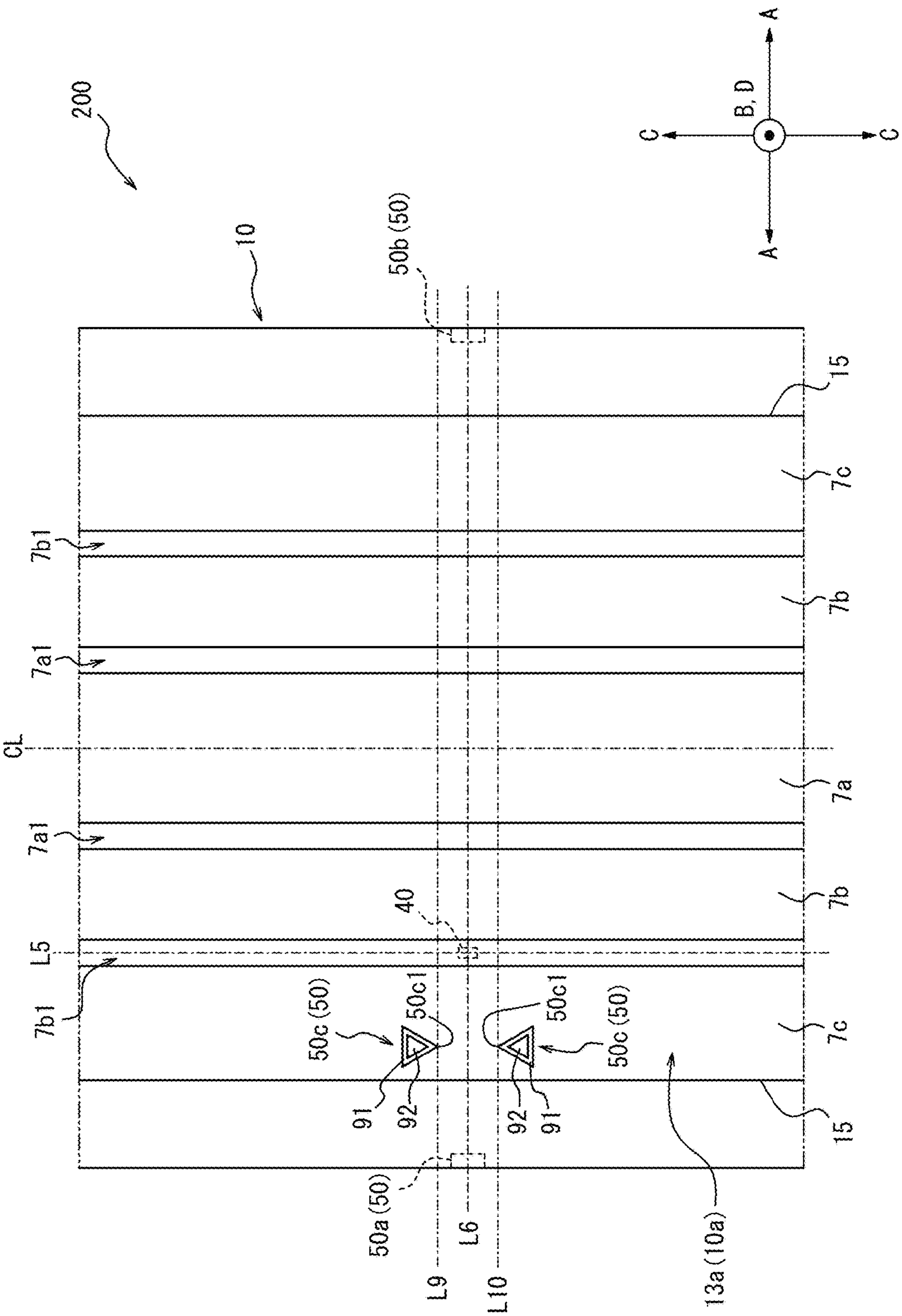
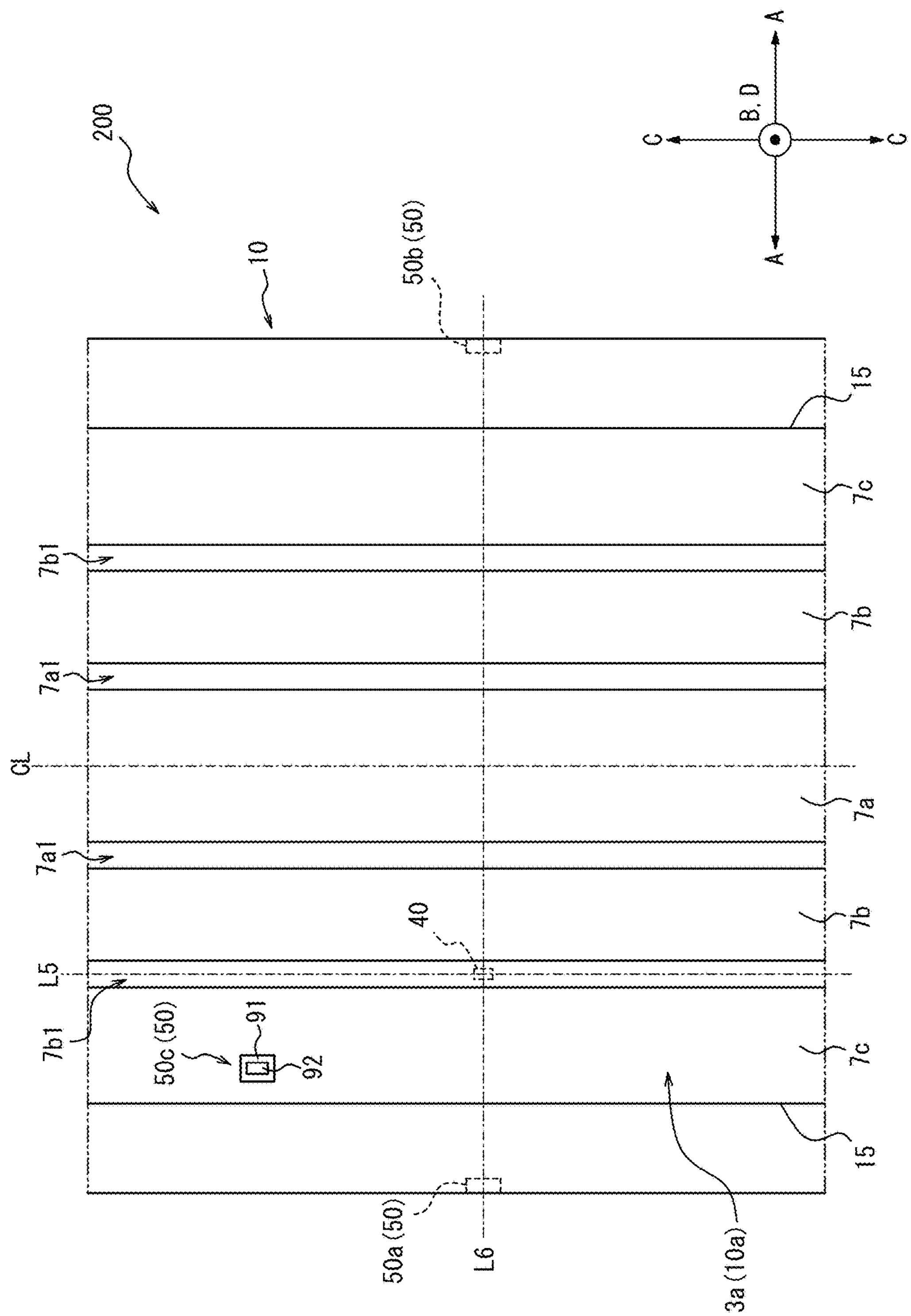

200
10
50b (50)
15
7c
7b1
7b
7a1
CL
7a
7a1
7b
40
L5
7b1
50c (50)
50c1
91
92
50c1
50c (50)
92
91
7c
15
50a (50)
L6
L9
L10
13a (10a)
A
A
C
C
B, D

FIG. 8E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10
50b (50)
A
B, D
C
C
A
15
7c
7b1
7b
7a1
CL
7a
7a1
7b
40
L5
7b1
50c (50)
91
92
7c
15
50a (50)
L6
13a (10a)

# TIRE

## TECHNICAL FIELD

The present disclosure relates to a tire.

## BACKGROUND

Tires with communication devices such as RF tags embedded within are known. Patent Literature (PTL) 1 discloses this type of tire.

## CITATION LIST

### Patent Literature

PTL 1: JP 2004-148953 A

## SUMMARY

### Technical Problem

It is difficult to distinguish the position of a communication device embedded within a tire or mounted on an inner surface of a tire from outside the tire at night or in a dark surrounding environment.

It would be helpful to provide a tire in which the position of a communication device can be distinguished from outside the tire in a dark surrounding environment.

### Solution to Problem

A tire, as a first aspect of the present disclosure, includes a tire body and a communication device embedded within the tire body or mounted on an inner surface of the tire body. The tire includes a label area that is located on an outer surface of the tire body and indicates the position of the communication device. The label area includes a light emission area.

### Advantageous Effect

According to the present disclosure, it is possible to provide a tire in which the position of a communication device can be distinguished from outside the tire in a dark surrounding environment.

## BRIEF DESCRIPTION OF THE DRAWINGS

In the accompanying drawings:

FIG. **1** is a cross-sectional view of a tire as an embodiment of the present disclosure in a tire width direction;

FIG. **2** is a front view of a first label area of the tire illustrated in FIG. **1**;

FIG. **3** is a side view of the tire illustrated in FIG. **1**, viewed from the side of a first side portion;

FIG. **4**A is a diagram illustrating a variation of the first label area illustrated in FIG. **1**;

FIG. **4**B is a diagram illustrating a variation of the first label area illustrated in FIG. **1**;

FIG. **4**C is a diagram illustrating a variation of the first label area illustrated in FIG. **1**;

FIG. **4**D is a diagram illustrating a variation of the first label area illustrated in FIG. **1**;

FIG. **4**E is a diagram illustrating a variation of the first label area illustrated in FIG. **1**;

FIG. **5** is a cross-sectional view of a tire as an embodiment of the present disclosure in the tire width direction;

FIG. **6** is a cross-sectional view of a tire as an embodiment of the present disclosure in the tire width direction;

FIG. **7** is a part of a development diagram of a tread outer surface of the tire illustrated in FIG. **6**;

FIG. **8**A is a diagram illustrating a variation of a third label area illustrated in FIG. **6**;

FIG. **8**B is a diagram illustrating a variation of the third label area illustrated in FIG. **6**;

FIG. **8**C is a diagram illustrating a variation of the third label area illustrated in FIG. **6**;

FIG. **8**D is a diagram illustrating a variation of the third label area illustrated in FIG. **6**; and

FIG. **8**E is a diagram illustrating a variation of the third label area illustrated in FIG. **6**.

## DETAILED DESCRIPTION

Embodiments of a tire according to the present disclosure will be exemplarily described below with reference to the drawings. In the drawings, the same reference signs denote common components. In this specification, a tire width direction refers to a direction parallel to the central axis of the tire. A tire radial direction refers to a radial direction that is orthogonal to the central axis of the tire and centered on the central axis. A tire circumferential direction refers to a direction in which the tire rotates about the central axis of the tire.

In this specification, a "tread outer surface" means an outer circumferential surface throughout the entire circumference of the tire that is in contact with a road surface when the tire mounted on a rim and filled with a specified internal pressure is rotated under a maximum load (hereinafter also referred to as "maximum load condition"). In addition, "tread ends" mean outer ends of the tread outer surface in the tire width direction.

In this specification, the "rim" means a standard rim (Measuring Rim in ETRTO's STANDARDS MANUAL and Design Rim in TRA's YEAR BOOK) of an applicable size conformable to industrial standards valid for regions in which tires are produced and used, as described in or to be described in JATMA YEAR BOOK of the Japan Automobile Tyre Manufacturers Association, Inc. (JATMA) in Japan, STANDARDS MANUAL of the European Tyre and Rim Technical Organisation (ETRTO) in Europe, YEAR BOOK of the Tire and Rim Association, Inc. (TRA) in the United States, or the like, but in the case of a size that is not listed in such industrial standards, a rim with a width corresponding to a bead width of the tire. The "rim" may be of a size that may be included in the above industrial standards in the future, as well as a current size. Examples of the "size that is included in the future" may be sizes listed as "FUTURE DEVELOPMENTS" in the 2013 edition of STANDARDS MANUAL of the ETRTO.

In this specification, the "specified internal pressure" means an air pressure (maximum air pressure) corresponding to a maximum load capacity of a single wheel in an applicable size and ply rating as described in the above industrial standards such as JATMA YEAR BOOK, but in the case of a size that is not listed in the above industry standards, an air pressure (maximum air pressure) corresponding to a maximum load capacity specified for each vehicle on which the tire is mounted. In this specification, the "maximum load" means a load corresponding to the maximum load capacity of the tire of an applicable size listed in the above industrial standards, but in the case of a size that is not listed in the above industrial standards, a load corresponding to the maximum load capacity specified for each vehicle on which the tire is mounted.

## FIRST EMBODIMENT

A pneumatic tire **1** (hereinafter simply referred to as "tire **1**") as an embodiment of the tire according to the present disclosure will be exemplarily described below with reference to the drawings. Unless otherwise stated, the tire **1** described below is assumed to mean a tire **1** that is mounted on a rim, filled with a specified internal pressure, and in a maximum load condition. The type of tire **1** is not particularly limited, and may be, for example, a tire for trucks or buses, or a tire for passenger cars.

FIG. **1** is a cross-sectional view of the tire **1** in a tire width direction. As illustrated in FIG. **1**, the tire **1** includes a tire body **10** and a communication device **40**.

The tire body **10** includes a tread portion **10***a*, a first side portion **10***b***1**, and a second side portion **10***b***2**. The first side portion **10***b***1** extends from one end of the tread portion **10***a* in a tire width direction A toward an inner side B**2** in a tire radial direction B. The second side portion **10***b***2** extends from the other end of the tread portion **10***a* in the tire width direction A toward the inner side B**2** in the tire radial direction B. Hereinafter, the first side portion **10***b***1** and the second side portion **10***b***2** are simply described as "side portions **10***b*" when not specifically distinguished.

The side portions **10***b* each have a sidewall portion **11** and a bead portion **12**. The sidewall portion **11** is connected to the end of the tread portion **10***a* in the tire width direction A and extends toward the inner side B**2** in the tire radial direction B. The bead portion **12** is connected to an end of the sidewall portion **11** on the inner side B**2** in the tire radial direction B.

More specifically, the tire body **10** of this embodiment includes bead cores **2**, bead fillers **3**, a carcass **4**, belts **5**, tread rubber **7**, side rubber **8**, and an inner liner **9**. However, the tire body **10** may include other components, not limited to the above components.

[Tire Body **10**]

[[Bead Cores **2** and Bead Fillers **3**]]

The bead cores **2** and the bead fillers **3** are embedded in the bead portions **12** of the side portions **10***b*. Each bead core **2** includes a bead cord that is surrounded by a rubber coating. The bead cord is formed of a steel cord. Each bead filler **3** is made of rubber and is located on an outer side B**1** in the tire radial direction B with respect to the bead core **2**.

[[Carcass **4**]]

The carcass **4** extends toroidally between the pair of bead portions **12**, which is constituted of the bead portion **12** of the first side portion **10***b***1** and the bead portion **12** of the second side portion **10***b***2**, more specifically between the pair of bead cores **2**. Specifically, the carcass **4** of this embodiment includes a carcass ply **4***a*. The carcass ply **4***a* is folded from the inside to the outside in the tire width direction A around the respective bead cores **2**. The carcass ply **4***a* may have a plurality of ply cords arranged parallel to each other and coating rubber that coats the plurality of ply cords. The carcass **4** of this embodiment includes only one carcass ply **4***a*, but may include two or more carcass plies **4***a*. The plurality of ply cords of the carcass ply **4***a* may be arranged at an angle of, for example, 75° to 90° with respect to a tire circumferential direction C. The ply cords of the carcass ply **4***a* can be, for example, metal cords such as steel cords.

More specifically, the carcass ply **4***a* of this embodiment has a body part **4***a***1** located between the pair of bead cores **2**, and folded parts **4***a***2** that are connected to the body part **4***a***1** and folded from the inside to the outside in the tire width direction A around the respective bead cores **2**. The above-described bead fillers **3** are each located between the body part **4***a***1** and each folded part **4***a***2** of the carcass ply **4***a*.

[[Belts **5**]]

The belts **5** are embedded in the tread portion **10***a*. The tire body **10** of this embodiment has four of the belts **5** on the outer side B**1** of a crown part of the carcass **4** in the tire radial direction B. The four belts **5** of this embodiment include a first belt **5***a*, a second belt **5***b*, a third belt **5***c*, and a fourth belt **5***d*. Hereafter, the first belt **5***a*, the second belt **5***b*, the third belt **5***c*, and the fourth belt **5***d* are simply referred to as "belts **5**" when not specifically distinguished. The first belt **5***a* is located at the most inner side B**2** in the tire radial direction B, among the first belt **5***a*, the second belt **5***b*, the third belt **5***c*, and the fourth belt **5***d*. The fourth belt **5***d* is located at the most outer side B**1** in the tire radial direction B, among the first belt **5***a*, the second belt **5***b*, the third belt **5***c*, and the fourth belt **5***d*. The second belt **5***b* and the third belt **5***c* are located between the first belt **5***a* and the fourth belt **5***d* in the tire radial direction B. More specifically, the third belt **5***c* is located on the outer side B**1** of the second belt **5***b* in the tire radial direction B.

In this embodiment, the first belt **5***a*, the second belt **5***b*, the third belt **5***c*, and the fourth belt **5***d* constitute four belt layers. Specifically, the first belt **5***a* forms a first belt layer **20***a* located at the most inner side B**2** in the tire radial direction B. The fourth belt **5***d* forms a fourth belt layer **20***d* located at the most outer side B**1** in the tire radial direction B. The second belt **5***b* forms a second belt layer **20***b*. The third belt **5***c* forms a third belt layer **20***c*.

Each belt **5** may include a plurality of belt cords arranged in parallel to each other and coating rubber that coats the plurality of belt cords. The first to fourth belt layers **20***a* to **20***d* are inclined belt layers in which the belt cords form a predetermined angle with respect to the tire circumferential direction C. The belt cords of two belt layers adjacent in the tire radial direction B, among the first to fourth belt layers **20***a* to **20***d*, may be inclined in the same direction or in opposite directions with respect to the tire circumferential direction C. The material of the belt cords of the belts **5** is not particularly limited.

The belt cords of the belts **5** may be, for example, metal cords or organic fiber cords. A rubber composition used for the coating rubber of the belts **5** is not particularly limited.

[[Tread Rubber **7**]]

As illustrated in FIG. **1**, the tread rubber **7** is located on the outer side B**1** of the crown part of the body part **4***a***1** of the carcass ply **4***a* and on the outer side B**1** of the first to fourth belts **5***a* to **5***d*, which constitute the first to fourth belt layers **20***a* to **20***d*, in the tire radial direction B. A tread outer surface **13***a*, which is an outer surface of the tread portion **10***a* of this embodiment, is composed of the tread rubber **7**. In the tread outer surface **13***a*, as illustrated in FIG. **1**, there are formed two inner circumferential grooves **7***a***1**, which zone a center land area **7***a* through which a tire equatorial plane CL passes, and two outer circumferential grooves **7***b***1**, which are located outside the inner circumferential grooves **7***a***1** in the tire width direction A and zone intermediate land areas **7***b* each between the inner circumferential groove **7***a***1** and the outer circumferential groove **7***b***1**. In the tread outer surface **13***a* of this embodiment, shoulder land areas **7***c* are zoned each between the outer circumferential groove **7***b***1** and a tread end **15** located outside the outer circumferential groove **7***b***1** in the tire width direction A. In the tread outer surface **13***a*, for example, width directional grooves extending in the tire width direction A or the like may be formed.

[[Side Rubber **8**]]

The side rubber **8** is disposed outside the body part **4***a***1** and the folded parts **4***a***2** of the carcass ply **4***a* in the tire width direction A. Side outer surfaces **14***a*, which are outer surfaces of the side portions **10***b* of this embodiment, are composed of the side rubber **8**. Ends of the side rubber **8** on the outer side B**1** in the tire radial direction B are connected to ends of the above-described tread rubber **7** in the tire width direction A.

Thus, in this embodiment, an outer surface of the tire body **10** includes the tread outer surface **13***a* and the side outer surfaces **14***a*, which are connected to the tread outer surface **13***a*. The tire body **10** includes, on its outer surface, label areas **50** that indicate the position of a communication device **40** described below. The details of the distinction areas **50** will be described below (see FIGS. **2**, **3**, and the like).

[[Inner Liner **9**]]

The inner liner **9** covers a tire inner surface side of the body part **4***a***1** of the carcass ply **4***a*, and constitutes an inner surface of the tire body **10**. The inner surface of the tire body **10** includes a tread inner surface **13***b*, which is an inner surface of the tread portion **10***a*, and side inner surfaces **14***b*, which are inner surfaces of the side portions **10***b*. The inner liner **9** is laminated on the tire inner surface side of the body part **4***a***1** of the carcass ply **4***a*. The inner liner **9** may be formed of, for example, butyl rubber with low air permeability.

[Communication Device **40**]

As illustrated in FIG. **1**, the communication device **40** is embedded within the tire body **10**. However, the communication device **40** may be mounted on the inner surface of the tire body **10**. The inner surface of the tire body **10** on which the communication device **40** is mounted may be the tread inner surface **13***b*, a first side inner surface **14***b***1**, which is an inner surface of the first side portion **10***b***1**, or a second side inner surface **14***b***2**, which is an inner surface of the second side portion **10***b***2**.

As illustrated in FIG. **1**, the communication device **40** of this embodiment is embedded within the side portion **10***b* of the tire body **10**. More specifically, the communication device **40** of this embodiment is embedded within the first side portion **10***b***1** of the tire body **10**. The communication device **40** of this embodiment is embedded, in the first side portion **10***b***1**, between the body part **4***a***1** of the carcass **4** and the side rubber **8**. However, the communication device **40** may be embedded in another location in the first side portion **10***b***1**. As described above, the communication device **40** may also be mounted on the inner surface of the first side portion **10***b***1** (see the position represented by the alternate long and short dashed line with the reference sign "**40**" in FIG. **1**).

Furthermore, the communication device **40** may be disposed in another location than the first side portion **10***b***1**. The communication device **40** may be, for example, embedded within the second side portion **10***b***2** of the tire body **10**, or mounted on the inner surface of the second side portion **10***b***2**. The communication device **40** may be embedded within the tread portion **10***a* of the tire body **10** (see FIG. **5**), or may be mounted on the inner surface of the tread portion **10***a* (see FIG. **6**).

The configuration of the communication device **40** is not particularly limited as long as the communication device **40** is configured to be able to wirelessly communicate with a predetermined device outside the tire body **10**. An example of the communication device **40** is an RF tag. The RF tag, as the communication device **40**, can wirelessly communicate with a reader/writer located outside the tire body **10**. The RF tag may be, for example, a passive RF tag that is operated by electric power supplied from the reader/writer located outside the tire body **10**. Specifically, the RF tag, as the communication device **40**, can receive, at an antenna of the RF tag, information transmitted from an antenna of the reader/writer in the form of a radio wave or a magnetic field. Due to rectification (in the case of the radio wave) or resonance (in the case of the magnetic field), electric power is generated at the antenna of the RF tag, and a memory and a controller of the RF tag perform predetermined operations. The controller of the RF tag can, for example, read out information from the memory of the RF tag and transmit the information (back) to the reader/writer from the antenna in the form of a radio wave or a magnetic field. The antenna of the reader/writer receives the radio wave or the magnetic field from the RF tag. A controller of the reader/writer retrieves the received information to acquire the information stored in the memory of the RF tag. The above-described memory and controller of the RF tag may be configured, for example, in an integrated circuit (IC chip) including a non-volatile memory.



The communication device **40** may be embedded within the tire body **10**, for example, as a covered communication device whose periphery is covered in advance with covering rubber. In such a case, the covering rubber that covers the periphery of the communication device **40** may be colored differently from adjacent portions of the tire body **10**.

[[[Label Areas **50** on Outer Surfaces of Tire Body **10**]]]

As illustrated in FIG. **1**, the label areas **50** indicating the position of the communication device **40** are provided on the outer surfaces of the tire body **10**. The label areas **50** of this embodiment are provided on the side outer surfaces **14***a*, which are the outer surfaces of the side portions **10***b* of the tire body **10**. More specifically, the tire body **10** of this embodiment includes a first label area **50***a* provided on a first side outer surface **14***a***1**, which is an outer surface of the first side portion **10***b***1**, and a second label area **50***b* provided on a second side outer surface **14***a***2**, which is an outer surface of the second side portion **10***b***2**.

FIGS. **2** and **3** are diagrams illustrating the first label area **50***a* from the front. In particular, FIG. **3** is a side view of the tire **1** viewed from the side of the first side portion **10***b***1**. In FIG. **3**, the position of the communication device **40** is represented by a dashed line.

As illustrated in FIG. **3**, in this embodiment, the first label area **50***a* and the communication device **40** are disposed at the same position in the tire radial direction B. Here, "the first label area **50***a* and the communication device **40** are disposed at the same position in the tire radial direction B" means that there is at least one virtual circle L**1** around the central axis O of the tire **1** that passes through the first label area **50***a* and the communication device **40** in the side view of the tire **1** (see FIG. **3**). In other words, it is not limited to a case in which the extension areas of the respective first label area **50***a* and communication device **40** in the tire radial direction B are completely coincident.

As illustrated in FIG. **3**, in this embodiment, the first label area **50***a* and the communication device **40** are disposed at the same position in the tire circumferential direction C. Here, "the first label area **50***a* and the communication device **40** are disposed at the same position in the tire circumferential direction C" means that, in the side view of the tire **1** (see FIG. **3**), there is at least one virtual straight line L**2** that passes through the central axis O of the tire **1** and passes through the first label area **50***a* and the communication device **40** on one side with respect to the central axis O. In other words, it is not limited to a case in which the extension areas of the respective first label area **50***a* and communication device **40** in the tire circumferential direction C are completely coincident.

In other words, the first label area **50***a* and the communication device **40** of this embodiment overlap in a tire wall thickness direction D. The communication device **40** of this embodiment overlaps with the first label area **50***a* in the tire wall thickness direction D in its entire extension areas in the tire radial direction B and the tire circumferential direction C. In other words, the entire communication device **40** of this embodiment overlaps with the first label area **50***a* in the tire wall thickness direction D. However, only a part of the communication device **40** may overlap with the first label area **50***a* in the tire wall thickness direction D. The entire first label area **50***a* may overlap with the communication device **40** in the tire wall thickness direction D. Furthermore, the contour of the first label area **50***a* may coincide with the contour of the communication device **40** in the tire wall thickness direction D. In other words, the entire first label area **50***a* may overlap with the entire communication device **40** in the tire wall thickness direction D.

Thus, the first label area **50***a* of this embodiment is disposed in such a position that at least a part of the first label area **50***a* overlaps with the communication device **40** in the tire wall thickness direction D. Therefore, the position of the communication device **40** can be identified by identifying the first label area **50***a* from outside the tire **1**.

As illustrated in FIGS. **1** to **3**, the first label area **50***a* of this embodiment includes a light emission area **61**. The light emission area **61** may be, for example, a phosphorescent area that emits stored light. The provision of the light emission area **61** in the first label area **50***a* allows the first label area **50***a* to be easily identified from outside the tire **1** in a dark environment, such as at night or in a dark place. Therefore, the position of the communication device **40** can be easily identified in the dark environment.

The light emission area **61** of the first label area **50***a* is not particularly limited as long as the light emission area **61** is an area that emits light to the outside. The light emission area **61** may be composed of, for example, phosphorescent paint containing a phosphorescent material. The light emission area **61** may be formed, for example, by mixing a phosphorescent material into a part of a rubber composition configuring the outer surface of the tire body **10**. The rubber composition containing the phosphorescent material may be, for example, affixed to an unvulcanized tire and vulcanized together with the unvulcanized tire. The rubber composition containing the phosphorescent material may also be, for example, affixed to a tire after vulcanization. Furthermore, the light emission area **61** may be formed, for example, by a resin phosphorescent material configuring a part of the outer surface of the tire. The resin phosphorescent material may be formed, for example, by carrying a phosphorescent material in void spaces of a foam resin, such as polyurethane foam. The resin phosphorescent material may be, for example, affixed to an unvulcanized tire and vulcanized together with the unvulcanized tire. The resin phosphorescent material may also be, for example, affixed to a tire after vulcanization.

The phosphorescent material is not particularly limited, and any known phosphorescent material may be used. Examples of the phosphorescent material include, for example, compositions: SrAl2O4: Eu, Dy; Sr4Al14O25: Eu, Dy; SrAl2O4: Eu, Dy+Sr4Al14O25: Eu, Dy; Sr4Al14O25: Eu, Dy+CaAl2O4: Eu, Nd; CaAl2O4: Eu, Nd; ZnS: Cu, Mn, Co; ZnS: Cu; and the like.

The light emission area **61** of the first label area **50***a* preferably has a color or shape that can be distinguished from the surroundings. As illustrated in FIG. **1**, the light emission area **61** of this embodiment is provided on a top surface of a convex portion **70**, which protrudes outward in the tire width direction A from the surroundings. Thus, the light emission area **61** has a color or shape that can be distinguished from the surroundings, so that the first label area **50***a* can be easily identified not only in a dark environment but also in a bright environment, for example, during the daytime. In other words, the position of the communication device **40** can be distinguished from outside the tire **1** even in the bright environment.

The light emission area **61** of this embodiment is provided on the top surface of the convex portion **70**, but is not limited to this configuration. The light emission area **61** may be provided, for example, on a bottom surface of a concave portion. In addition to or instead of being provided on the top surface of the convex portion **70** or the bottom surface of the concave portion, for example, the light emission area **61** may have a different color from the surroundings. Here, the "different color" means not only a difference in at least one of hue, brightness, or saturation, but also a difference in pattern, luster, or the like, such that the difference can be visually grasped. The coloring of the light emission area **61** can be achieved, for example, by adding pigment together with the phosphorescent material described above, or by using pigment containing the phosphorescent material described above. However, as in the present embodiment, the light emission area **61** is preferably provided at least on the top surface of the convex portion **70**. This prevents a reduction in the visibility of the light emission area **61** from the outside, for example, in a case in which the light emission area **61** is covered with mud or the like and is difficult to see from the outside. The light emission area **61** provided on the top surface of the convex portion **70** may be composed of, for example, phosphorescent paint laminated on the top surface of the convex portion **70**. The light emission area **61** provided on the top surface of the convex portion **70** may be realized, for example, by making the convex portion **70** itself of the rubber composition containing the phosphorescent material, the resin phosphorescent material, or the like as described above.

As illustrated in FIG. **2**, the first label area **50***a* of this embodiment preferably includes a non-light emission area **62** surrounded by the light emission area **61**. The provision of such a non-light emission area **62** emphasizes the contrast between the light emission area **61** and the non-light emission area **62** in a dark environment. Therefore, the visibility of the first label area **50***a* in the dark environment can be enhanced.

In this embodiment, the entire circumference of the non-light emission area **62** is surrounded by the light emission area **61** in the front view of the first label area **50***a* (see FIGS. **2** and **3**). More specifically, the light emission area **61** of the first label area **50***a* of this embodiment is annular in shape in the front view of the first label area **50***a* (see FIGS. **2** and **3**). In other words, the convex portion **70** of this embodiment is annular in shape in the front view of the first label area **50***a* (see FIGS. **2** and **3**), and the light emission area **61** is provided on the top surface of this annular convex portion **70**. The non-light emission area **62** is provided inside the annular light emission area **61** in the front view of the first label area **50***a* (see FIGS. **2** and **3**). However, the configuration of the light emission area **61** and the non-light emission area **62** of the first label area **50***a* is not limited to this. As long as the non-light emission area **62** of the first label area **50***a* can be identified in a dark environment, there may be a position at which no light emission area **61** is provided around the non-light emission area **62**. Such an aspect includes, for example, a non-light emission area **62** surrounded by an approximately C-shaped light emission area **61** in the front view of the first label area **50***a* (see FIGS. **2** and **3**). However, as in this embodiment, the entire circumference of the non-light emission area **62** is preferably surrounded by the light emission area **61** in the front view of the first label area **50***a* (see FIGS. **2** and **3**). This can further emphasize the contrast between the light emission area **61** and the non-light emission area **62** in a dark environment.

Furthermore, in this embodiment, only the non-light emission area **62** of the first label area **50***a* is configured to overlap with the communication device **40** in the tire wall thickness direction D. In other words, the light emission area **61** of the first label area **50***a* is configured not to overlap with the communication device **40** in the tire wall thickness direction D. Thereby, the position outside the contour (see the dashed line in FIG. **3**) of the communication device **40** can be identified by the light emission area **61** in the front view of the first label area **50***a* (see FIGS. **2** and **3**). Therefore, the entire communication device **40** can be easily removed from the tire body **10** by using the first label area **50***a*, for example, at the time of disposal of the tire **1**.

The non-light emission area **62** may have a different color from the light emission area **61**. The non-light emission area **62** may have a different color from the color of the first side outer surface **14***a***1** of the first side portion **10***b***1** located around the first label area **50***a*. This can enhance the distinguishability of each of the light emission area **61**, the non-light emission area **62**, and the surroundings of the first label area **50***a* in a bright environment.

The non-light emission area **62** of a different color from the light emission area **61** may be configured, for example, by paint containing pigment. The non-light emission area **62** of a different color from the light emission area **61** may be formed, for example, by coloring the non-light emission area **62** with a mixture of other pigment than carbon black into a part of the rubber composition composing the outer surface of the tire body **10**. Such a colored rubber composition may be, for example, affixed to an unvulcanized tire and vulcanized together with the unvulcanized tire. The colored rubber composition may also be affixed to a tire after vulcanization, for example. Furthermore, the non-light emission area **62** of a different color from the light emission area **61** may be formed of, for example, a resin coloring body that constitutes a part of the outer surface of the tire. The resin coloring body may be formed, for example, by carrying pigment in void spaces of a foam resin, such as polyurethane foam. The resin coloring body may be, for example, affixed to an unvulcanized tire and vulcanized together with the unvulcanized tire. The resin coloring body may also be affixed to a tire after vulcanization, for example.

As illustrated in FIG. **1**, the non-light emission area **62** may be provided on a fine uneven surface. The fine uneven surface may have, for example, a base surface and a plurality of ridges **68** with a height of 0.1 to 1.0 mm that is arranged in parallel on the base surface. The distance between the tops of two adjacent ridges **68** is preferably 0.3 to 1.5 mm, for example. The provision of the non-light emission area **62** on such a fine uneven surface can control the reflection of light from the non-light emission area **62** in a bright environment, and therefore can achieve a high contrast with the surrounding light emission area **61**, which reflects light. Therefore, the distinguishability of each of the light emission area **61** and the non-light emission area **62** can be further enhanced in the bright environment.

The tire body **10** of this embodiment includes a cylindrical base protrusion portion **69**, which protrudes on the outer surface of the first side portion **10***b***1**, and the annular convex portion **70**, which protrudes from an outer edge of a top surface of the base protrusion portion **69**. The first label area **50***a* of this embodiment is constituted of the light emission area **61**, which is provided on the top surface of the convex portion **70**, and the non-light emission area **62**, which is provided on the top surface of the base protrusion portion **69** inside the convex portion **70**. In other words, in this embodiment, not only the light emission area **61** but also the non-light emission area **62** is provided on the top surface of the portion protruding from the surroundings of the first label area **50***a*. By providing the entire first label area **50***a* on the top surface of the portion protruding on the outer surface of the tire body **10**, as described above, the distinguishability of the first label area **50***a* can be further enhanced.

Such a first label area **50***a* can be achieved, for example, by laminating paint containing a phosphorescent material on the top surface of the convex portion **70** and laminating paint containing pigment on the top surface of the base protrusion portion **69** inside the convex portion **70**. A component corresponding to the base protrusion portion **69** and a component corresponding to the convex portion **70** may be formed separately and then laminated to form a single unit. The component corresponding to the base protrusion portion **69** may be formed of, for example, the above-described colored rubber composition, resin colored material, or the like. The component corresponding to the convex portion **70** may be formed of, for example, the above-described rubber composition containing the phosphorescent material, the resin phosphorescent material, or the like.

As illustrated in FIG. **1**, in this embodiment, the second label area **50***b* is provided on the second side outer surface **14***a***2** of the second side portion **10***b***2** of the tire body **10**. The second label area **50***b* is configured to be distinguishable from the first label area **50***a*. Specifically, the second label area **50***b* of this embodiment is constituted only of a light emission area **81**. In other words, the second label area **50***b* of this embodiment is the light emission area **81** in the entirety of its contour in the front view, and does not include a non-light emission area. By making the first label area **50***a* and the second label area **50***b* distinguishable, as described above, it is possible to prevent confusion between the first label area **50***a* and the second label area **50***b* and to distinguish between information indicated by the first label area **50***a* and information indicated by the second label area **50***b*.

In this embodiment, the information indicated by the first label area **50***a* is the position itself of the communication device **40** overlapping in the tire wall thickness direction D. On the contrary, the information indicated by the second label area **50***b* is the position of the communication device **40** in the tire circumferential direction C. The second label area **50***b* and the communication device **40** of this embodiment are disposed at the same position in the tire circumferential direction C. Here, "the second label area **50***b* and the communication device **40** are disposed at the same position in the tire circumferential direction C" is the same as the relationship described above in which the first label area **50***a* and the communication device **40** are disposed at the same position in the tire circumferential direction C. In other words, the second label area **50***b* of this embodiment indicates only the position of the communication device **40** in the tire circumferential direction C. The provision of the second label area **50***b*, in addition to the first label area **50***a*, allows the position of the communication device **40** in the tire circumferential direction C to be identified even from the side of the second side portion **10***b***2**, in contrast to the case of providing only the first label area **50***a*. Therefore, for example, even when the tire **1** is mounted on a vehicle with the first side portion **10***b***1** located inside the vehicle, and the first label area **50***a* is not visible from outside the vehicle, the position of the communication device **40** can be narrowed down based on the second label area **50***b* on the second side portion **10***b***2**. The position of the communication device **40** can be easily identified regardless of the orientation of the first side portion **10***b***1** of the tire body **10** mounted on the vehicle.

The second label area **50***b* of this embodiment indicates the position of the communication device **40** in the tire circumferential direction C, but is not limited to this configuration. The second label area **50***b* may indicate the position of the communication device **40** in the tire radial direction B. In other words, the second label area **50***b* and the communication device **40** may be disposed at the same position in the tire radial direction B. Here, "the second label area **50***b* and the communication device **40** are disposed at the same position in the tire radial direction B" is the same as the relationship described above in which the first label area **50***a* and the communication device **40** are disposed at the same position in the tire radial direction B. The second label area **50***b* may not indicate the position of the communication device **40** in the tire radial direction B and the tire circumferential direction C, but only indicate that the communication device **40** is in/on the first side portion **10***b***1**. In other words, the second label area **50***b* may be a label indicating that there is no communication device **40** in/on the second side portion **10***b***2**. Thus, the second label area **50***b* is not particularly limited as long as the second label area **50***b* indicates some location information on the communication device **40**. However, as described above, it is preferable that the second label area **50***b* indicates at least the position of the communication device **40** in the tire circumferential direction C.

The light emission area **81** of the second label area **50***b* preferably has a color or shape that can be distinguished from the surroundings. As illustrated in FIG. **1**, the light emission area **81** of this embodiment is provided on a top surface of a convex portion **71**, which protrudes outward in the tire width direction A from the surroundings. Thus, due to the light emission area **81** having the color or the shape that can be distinguished from the surroundings, the second label area **50***b* can be easily identified not only in a dark environment but also in a bright environment, for example, during the daytime.

The light emission area **81** of this embodiment is provided on the top surface of the convex portion **71**, but is not limited to this configuration. The light emission area **81** may be provided, for example, on a bottom surface of a concave portion. In addition to or instead of being provided on the top surface of the convex portion **71** or the bottom surface of the concave portion, for example, the light emission area **81** may have a different color from its surroundings. However, as in the present embodiment, the light emission area **81** is preferably provided at least on the top surface of the convex portion **71**. This prevents a reduction in the visibility of the light emission area **81** from the outside, for example, in a case in which the light emission area **81** is covered with mud or the like and is difficult to see from the outside. The light emission area **81** may be formed of, for example, paint containing a phosphorescent material, or the like, by the same method as for the light emission area **61** of the first label area **50***a* as described above.

The light emission area **81** of the second label area **50***b* of this embodiment is provided on the entire top surface of the cylindrical convex portion **71**, but the aspect thereof is not particularly limited as long as the light emission area **81** can be distinguished from the light emission area **61** of the first label area **50***a* in a dark environment. The second label area **50***b* is configured by the top surface of the cylindrical convex portion **71**, but the aspect thereof is not particularly limited as long as the second label area **50***b* can be distinguished from the first label area **50***a* in a bright environment. The aspect of the second label area **50***b* is not particularly limited as long as the second label area **50***b* can be distinguished from the first label area **50***a* in the dark and bright environments.

Next, with reference to FIGS. **4**A to **4**E, variations of the first label area **50***a* will be described. FIGS. **4**A to **4**E are side views of the tire **1** viewed from the side of the first side portion **10***b***1**. In FIGS. **4**A to **4**E, the position of the communication device **40** embedded in the first side portion **10***b***1** is represented by a dashed line.

The first label area **50***a* illustrated in FIG. **4**A indicates the position of the communication device **40** in the tire circumferential direction C. However, the first label area **50***a* illustrated in FIG. **4**A does not indicate the position of the communication device **40** in the tire radial direction B. In other words, the first label area **50***a* and the communication device **40** illustrated in FIG. **4**A are disposed at the same position in the tire circumferential direction C, but are not disposed at the same position in the tire radial direction B. According to the first label area **50***a* illustrated in FIG. **4**A, the position of the communication device **40** in the tire circumferential direction C can be identified.

The first label area **50***a* illustrated in FIG. **4**B indicates the position of the communication device **40** in the tire radial direction B. However, the first label area **50***a* illustrated in FIG. **4**B does not indicate the position of the communication device **40** in the tire circumferential direction C. In other words, the first label area **50***a* and the communication device **40** illustrated in FIG. **4**B are disposed at the same position in the tire radial direction B, but are not disposed at the same position in the tire circumferential direction C. According to the first label area **50***a* illustrated in FIG. **4**B, the position of the communication device **40** in the tire radial direction B can be identified.

The first label areas **50***a* illustrated in FIG. **4**C indicate the position of the communication device **40** in the tire circumferential direction C. However, the first label areas **50***a* illustrated in FIG. **4**C do not indicate the position of the communication device **40** in the tire radial direction B. More specifically, the first label areas **50***a* illustrated in FIG. **4**C indicate a predetermined range in the tire circumferential direction C. The first label areas **50***a* illustrated in FIG. **4**C indicate the range in the tire circumferential direction C, in which the central angle around the central axis O of the tire **1** is "θ". The communication device **40** is located within the range in the tire circumferential direction C indicated by the first label areas **50***a*. According to the first label areas **50***a* illustrated in FIG. **4**C, the range in the tire circumferential direction C, in which the communication device **40** is located, can be identified.

In the example illustrated in FIG. **4**C, two of the first label areas **50***a* disposed at different positions in the tire circumferential direction C indicate the range in the tire circumferential direction C, in which the communication device **40** is located. Each first label area **50***a* is approximately triangular in shape in the front view. The first label areas **50***a* are disposed such that a vertex **50***a***1** of one of the first label areas **50***a* faces a vertex **50***a***1** of the other first label area **50***a* in the tire circumferential direction C. Thus, when the two first label areas **50***a* indicate the range in the tire circumferential direction C, the respective first label areas **50***a* preferably include the vertices **50***a***1** facing each other in the tire circumferential direction C. The provision of such vertices **50***a***1** makes it easier to visually grasp the range in the tire circumferential direction C indicated by the two first label areas **50***a*. The external shape of each first label area **50***a* in the front view is not limited to the triangular shape illustrated in FIG. **4**C. However, the triangular shape illustrated in FIG. **4**C is preferable as the external shape of each first label area **50***a* in the front view. This makes it further easier to visually grasp the range in the tire circumferential direction C indicated by the two first label areas **50***a*.

The first label areas **50***a* illustrated in FIG. **4**D indicate the position of the communication device **40** in the tire radial direction B and in the tire circumferential direction C. More specifically, the first label areas **50***a* illustrated in FIG. **4**D are disposed at the same position as the communication device **40** in the tire circumferential direction C. The first label areas **50***a* illustrated in FIG. **4**D also indicate a predetermined range in the tire radial direction B. The communication device **40** is located within the range in the tire radial direction B indicated by the first label areas **50***a*. According to the first label areas **50***a* illustrated in FIG. **4**D, it is possible to identify the position of the communication device **40** in the tire circumferential direction C, as well as the range in the tire radial direction B in which the communication device **40** is located.

In the example illustrated in FIG. **4**D, two of the first label areas **50***a* disposed at different positions in the tire radial direction B indicate the range in the tire radial direction B, in which the communication device **40** is located. Each first label area **50***a* is approximately triangular in shape in the front view. The first label areas **50***a* are disposed such that a vertex **50***a***1** of one of the first label areas **50***a* faces a vertex **50***a***1** of the other first label area **50***a* in the tire radial direction B. Thus, when the two first label areas **50***a* indicate the range in the tire radial direction B, the respective first label areas **50***a* preferably include the vertices **50***a***1** facing each other in the tire radial direction B. The provision of such vertices **50***a***1** makes it easier to visually grasp the range (the range between virtual circles L**3** and L**4** represented by long and short dashed lines in FIG. **4**D) in the tire radial direction B indicated by the two first label areas **50***a*. The external shape of each first label area **50***a* in the front view is not limited to the triangular shape illustrated in FIG. **4**D. However, the triangular shape illustrated in FIG. **4**D is preferable as the external shape of each first label area **50***a* in the front view. This makes it further easier to visually grasp the range in the tire radial direction B indicated by the two first label areas **50***a*.

In FIG. **4**D, the two first label areas **50***a* are disposed at the same position as the communication device **40** in the tire circumferential direction C, but may be disposed at different positions. However, as illustrated in FIG. **4**D, the two first label areas **50***a* are preferably disposed at the same position as the communication device **40** in the tire circumferential direction C. This makes it possible to easily identify the position of the communication device **40** in the tire circumferential direction C, in addition to the position of the communication device **40** in the tire radial direction B.

The first label area **50***a* illustrated in FIG. **4**E does not indicate the position of the communication device **40** in the tire radial direction B and the tire circumferential direction C, but only indicates that the communication device **40** is located in/on the first side portion **10***b***1**. In other words, the first label area **50***a* illustrated in FIG. **4**E may only indicate that the communication device **40** is located in/on the first side portion **10***b***1**, among the first side portion **10***b***1** and the second side portion **10***b***2** (see FIG. **1**). However, as illustrated in FIGS. **1** to **3** and FIGS. **4**A to **4**D, the first label area **50***a* preferably indicates the position of the communication device **40**, which is provided in/on the first side portion **10***b***1**, in at least one of the tire radial direction B or the tire circumferential direction C.

## SECOND EMBODIMENT

Next, with reference to FIG. **5**, a pneumatic tire **100** (hereinafter simply referred to as "tire **100**") as an embodiment of the tire according to the present disclosure will be exemplarily described. FIG. **5** is a cross-sectional view of the tire **100** in the tire width direction. The tire **100** of this embodiment differs from the tire **1** (see FIG. **1**) described above only in the position of the communication device **40**. Therefore, the position of the communication device **40** and information indicated by the first and second label areas **50***a* and **50***b* will be described here, and a description of the configuration common to the tire **1** of the first embodiment is omitted.

As illustrated in FIG. **5**, the tire **100** includes the communication device **40** embedded in the tread portion **10***a* of the tire body **10**. The communication device **40** of this embodiment is embedded within the tread portion **10***a*, but may be mounted on the tread inner surface **13***b*. As illustrated in FIG. **5**, the communication device **40** is disposed on one side in the tire width direction A with respect to the tire equatorial plane CL. For convenience of explanation, one side in the tire width direction A with respect to the tire equatorial plane CL, on which the communication device **40** is disposed, is hereinafter simply referred to as "communication device **40** side in the tire width direction A". The other side in the tire width direction A with respect to the tire equatorial plane CL, on which the communication device **40** is not disposed, is simply referred to as "opposite side of the communication device **40** side in the tire width direction A".

The configuration of the first label area **50***a* of this embodiment is the same as that of the first embodiment described above, so a description is omitted here. However, in contrast to the first embodiment, the first label area **50***a* of this embodiment does not overlap with the communication device **40** in the tire wall thickness direction D. The first label area **50***a* of this embodiment indicates the communication device **40** side in the tire width direction A. In other words, due to the first label area **50***a*, it is possible to identify which side in the tire width direction A the communication device **40** is located with respect to the tire equatorial plane CL.

Furthermore, the first label area **50***a* of this embodiment is preferably disposed at the same position as the communication device **40** in the tire circumferential direction C. This makes it possible to identify, using the first label area **50***a*, the position of the communication device **40** in the tire circumferential direction C.

The configuration of the second label area **50***b* of this embodiment is the same as that of the first embodiment described above, so a description is omitted here. Also, as in the first embodiment, the second label area **50***b* of this embodiment indicates the opposite side of the communication device **40** side in the tire width direction A. In other words, due to the second label area **50***b*, it is possible to identify which side in the tire width direction A the communication device **40** is located with respect to the tire equatorial plane CL.

Furthermore, the second label area **50***b* of this embodiment is preferably disposed at the same position as the communication device **40** in the tire circumferential direction C. This makes it possible to identify, from the second label area **50***b*, the position of the communication device **40** in the tire circumferential direction C.

## THIRD EMBODIMENT

Next, with reference to FIGS. **6** and **7**, a pneumatic tire **200** (hereinafter simply referred to as "tire **200**") as an embodiment of the tire according to the present disclosure will be exemplarily described. FIG. **6** is a cross-sectional view of the tire **200** in the tire width direction. FIG. **7** is a part of a development diagram of the tread outer surface **13***a* of the tire **200**. In FIG. **7**, the position of the communication device **40** is represented by the dashed line. Compared to the tire **1** (see FIG. **1**) and the tire **100** (see FIG. **5**) described above, the tire **200** of this embodiment differs in its configuration in terms of the position of the communication device **40** and the provision of a third label area **50***c*. Therefore, the differences will be mainly described here, and a description of a configuration common to the tire **1** of the first embodiment and the tire **100** of the second embodiment is omitted.

As illustrated in FIG. **6**, the tire **200** includes the communication device **40** mounted on the tread inner surface **13***b* of the tire body **10**. As illustrated in FIG. **5**, the communication device **40** of this embodiment is disposed on one side in the tire width direction A with respect to the tire equatorial plane CL. For convenience of explanation, one side in the tire width direction A with respect to the tire equatorial plane CL, on which the communication device **40** is disposed, is hereinafter simply referred to as "communication device **40** side in the tire width direction A". The other side in the tire width direction A with respect to the tire equatorial plane CL, on which the communication device **40** is not disposed, is simply referred to as "opposite side of the communication device **40** side in the tire width direction A".

The configuration of the first and second label areas **50***a* and **50***b* of this embodiment is the same as that of the first and second embodiments described above, so a description is omitted here. Information indicated by the first and second label areas **50***a* and **50***b* of this embodiment is also the same as that of the second embodiment described above, so a description is omitted here. However, the tire body **10** of the tire **200** of this embodiment may not include any one or both of the first label area **50***a* and the second label area **50***b*. The provision of the first label area **50***a* and the second label area **50***b* in the tire body **10** of the tire **200** allows the position of the communication device **40** to be identified more easily.

The third label area **50***c* is disposed on the tread outer surface **13***a*, which is an outer surface of the tread portion **10***a* of the tire body **10**.

As illustrated in FIG. **7**, in this embodiment, the third label area **50***c* and the communication device **40** are disposed at the same position in the tire width direction A. Here, "the third label area **50***c* and the communication device **40** are disposed at the same position in the tire width direction A" means that there is at least one virtual line L**5** that is in parallel to the tire circumferential direction C and passes through the third label area **50***c* and the communication device **40** in the developed view (see FIG. **7**) of the tread outer surface **13***a* of the tire **200**. In other words, it is not limited to a case in which the extension areas of the respective third label area **50***c* and communication device **40** in the tire width direction A are completely coincident.

As illustrated in FIG. **7**, in this embodiment, the third label area **50***c* and the communication device **40** are disposed at the same position in the tire circumferential direction C. Here, "the third label area **50***c* and the communication device **40** are disposed at the same position in the tire circumferential direction C" means that there is at least one virtual line L**6** that is in parallel to the tire width direction A and passes through the third label area **50***c* and the communication device **40** in the developed view (see FIG. **7**) of the tread outer surface **13***a* of the tire **200**. In other words, it is not limited to a case in which the extension areas of the respective third label area **50***c* and communication device **40** in the tire circumferential direction C are completely coincident.

In other words, the third label area **50***c* and the communication device **40** of this embodiment overlap in the tire wall thickness direction D. The communication device **40** overlaps with the third label area **50***c* in the tire wall thickness direction D in its entire extension areas in the tire width direction A and in the tire circumferential direction C. In other words, the entire communication device **40** of this embodiment overlaps with the third label area **50***c* in the tire wall thickness direction D. However, only a part of the communication device **40** may overlap with the third label area **50***c* in the tire wall thickness direction D. The entire third label area **50***c* may overlap with the communication device **40** in the tire wall thickness direction D. Furthermore, the contour of the third label area **50***c* may coincide with the contour of the communication device **40** in the tire wall thickness direction D. In other words, the entire third label area **50***c* may overlap with the entire communication device **40** in the tire wall thickness direction D.

Thus, the third label area **50***c* of this embodiment is disposed in such a position that at least a part of the third label area **50***c* overlaps with the communication device **40** in the tire wall thickness direction D. Therefore, the position of the communication device **40** can be identified by identifying the third label area **50***c* from outside the tire **200**.

As with the first label area **50***a* described above, the third label area **50***c* of this embodiment includes a light emission area **91**. The details of the light emission area **91** are similar to those of the light emission area **61** of the first label area **50***a* described above, except that the outer shape of the third label area **50***c* is approximately rectangular in the front view (see FIG. **7**), and that the third label area **50***c* is formed on a bottom surface of the outer circumferential groove **7***b***1**, so a description is omitted here. For distinguishability in a bright environment, the light emission area **91** of the third label area **50***c* may be provided, for example, on a top surface of a convex portion provided on the bottom surface of the outer circumferential groove **7***b***1**. The light emission area **91** of the third label area **50***c* may have, for example, a color that can be distinguished from the surroundings at the bottom surface of the outer circumferential groove **7***b***1**.

As illustrated in FIG. **7**, the third label area **50***c* of this embodiment preferably includes a non-light emission area **92** surrounded by the light emission area **91**. The details of the non-light emission area **92** are similar to those of the non-light emission area **62** of the first label area **50***a* described above, except that the outer shape of the third label area **50***c* is approximately rectangular in the front view (see FIG. 7), and that the third label area **50***c* is formed on the bottom surface of the outer circumferential groove 7*b***1**, so a description is omitted here.

Next, variations of the third label area **50***c* will be described with reference to FIGS. **8**A to **8**E. FIGS. **8**A to **8**E are each a part of a development diagram of the tread outer surface **13***a* of the tire **200**. In FIGS. **8**A to **8**E, the position of the communication device **40** mounted on the tread inner surface **13***b* is represented by a dashed line.

The third label area **50***c* illustrated in FIG. **8**A indicates the position of the communication device **40** in the tire width direction A. However, the third label area **50***c* illustrated in FIG. **8**A does not indicate the position of the communication device **40** in the tire circumferential direction C. In other words, the third label area **50***c* and the communication device **40** illustrated in FIG. **8**A are disposed at the same position in the tire width direction A, but are not disposed at the same position in the tire circumferential direction C. According to the third label area **50***c* illustrated in FIG. **8**A, the position of the communication device **40** in the tire width direction A can be identified.

The third label area **50***c* illustrated in FIG. **8**B indicates the position of the communication device **40** in the tire circumferential direction C. However, the third label area **50***c* illustrated in FIG. **8**B does not indicate the position of the communication device **40** in the tire width direction A. In other words, the third label area **50***c* and the communication device **40** illustrated in FIG. **8**B are disposed at the same position in the tire circumferential direction C, but are not disposed at the same position in the tire width direction A. According to the third label area **50***c* illustrated in FIG. **8**B, the position of the communication device **40** in the tire circumferential direction C can be identified.

The third label areas **50***c* illustrated in FIG. **8**C indicate the position of the communication device **40** in the tire width direction A. However, the third label areas **50***c* illustrated in FIG. **8**C do not indicate the position of the communication device **40** in the tire circumferential direction C. More specifically, the third label areas **50***c* illustrated in FIG. **8**C indicate a predetermined range in the tire width direction A. The communication device **40** is located within the range in the tire width direction A indicated by the third label areas **50***c*. According to the third label areas **50***c* illustrated in FIG. **8**C, the range in the tire width direction A, in which the communication device **40** is located, can be identified.

In the example illustrated in FIG. **8**C, two of the third label areas **50***c* disposed at different positions in the tire width direction A indicate the range in the tire width direction A, in which the communication device **40** is located. Each third label area **50***c* is approximately triangular in shape in the front view (see FIG. **7**). The third label areas **50***c* are disposed such that a vertex **50***cl* of one of the third label areas **50***c* faces a vertex **50***cl* of the other third label area **50***c* in the tire width direction A. Thus, when the two third label areas **50***c* indicate the range in the tire width direction A, the respective third label areas **50***c* preferably include the vertices **50***cl* facing each other in the tire width direction A. The provision of such vertices **50***cl* makes it easier to visually grasp the range in the tire width direction A indicated by the two third label areas **50***c*. The external shape of each third label area **50***c* in the front view is not limited to the triangular shape illustrated in FIG. **8**C. However, the triangular shape illustrated in FIG. **8**C is preferable as the external shape of each third label area **50***c* in the front view. This makes it further easier to visually grasp the range (the range between virtual straight lines L**7** and L**8** represented by long and short dashed lines in FIG. **8**C) in the tire width direction A indicated by the two third label areas **50***c*.

The third label areas **50***c* illustrated in FIG. **8**D indicate the position of the communication device **40** in the tire circumferential direction C. However, the third label areas **50***c* illustrated in FIG. **8**D do not indicate the position of the communication device **40** in the tire width direction A. More specifically, the third label areas **50***c* illustrated in FIG. **8**D indicate a predetermined range in the tire circumferential direction C. The communication device **40** is located within the range in the tire circumferential direction C indicated by the third label areas **50***c*. According to the third label areas **50***c* illustrated in FIG. **8**D, the range in the tire circumferential direction C, in which the communication device **40** is located, can be identified.

In the example illustrated in FIG. **8**D, the two third label areas **50***c* disposed at different positions in the tire circumferential direction C indicate the range in the tire circumferential direction C, in which the communication device **40** is located. Each third label area **50***c* is approximately triangular in shape in the front view (see FIG. **7**). The third label areas **50***c* are disposed such that a vertex **50***cl* of one of the third label areas **50***c* faces a vertex **50***cl* of the other third label area **50***c* in the tire circumferential direction C. Thus, when the two third label areas **50***c* indicate the range in the tire circumferential direction C, the respective third label areas **50***c* preferably include the vertices **50***cl* facing each other in the tire circumferential direction C. The provision of such vertices **50***cl* makes it easier to visually grasp the range in the tire circumferential direction C indicated by the two third label areas **50***c*. The external shape of each third label area **50***c* in the front view is not limited to the triangular shape illustrated in FIG. **8**D. However, the triangular shape illustrated in FIG. **8**D is preferable as the external shape of each third label area **50***c* in the front view. This makes it further easier to visually grasp the range (the range between virtual straight lines L**9** and L**10** represented by long and short dashed lines in FIG. **8**D) in the tire circumferential direction C indicated by the two third label areas **50***c*.

In FIG. **8**D, the two third label areas **50***c* are disposed at different positions from the communication device **40** in the tire width direction A, but may be disposed at the same position. This makes it possible to easily identify the position of the communication device **40** in the tire width direction A, in addition to the position of the communication device **40** in the tire circumferential direction C.

The third label area **50***c* illustrated in FIG. **8**E does not indicate the position of the communication device **40** in the tire width direction A and the tire circumferential direction C, but indicates the communication device **40** side of the tread portion **10***a* in the tire width direction A. In other words, the third label area **50***c* illustrated in FIG. **8**E may indicate one side of the tread portion **10***a* in the tire width direction A, on which the communication device **40** is disposed, with respect to the tire equatorial plane CL. However, as illustrated in FIGS. **6**, **7**, and **8**A to **8**D, the third label area **50***c* preferably indicates the position of the communication device **40** provided in the tread portion **10***a* in at least one of the tire radial direction B or the tire circumferential direction C.

As described above, due to the light emission areas of the various label areas **50** described in the first to third embodiments and the variations thereof, the position of the communication device **40** can be easily identified even in a dark environment.

In particular, the light emission area of the label area **50** is preferably disposed at the same position as the communication device **40** in at least one of the tire width direction A, the tire radial direction B, or the tire circumferential direction C. This allows the position of the communication device **40** to be identified more easily in a dark environment based on the light emission area of the label area **50**. In this specification, FIGS. **6**, **7**, and **8**A illustrate the examples in which the light emission area **91** of the third label area **50***c* is disposed at the same position as the communication device **40** in the tire width direction A. In this specification, FIGS. **1** to **3** and **4**B illustrate the examples in which the light emission area **61** of the first label area **50***a* and the light emission area **81** of the second label area **50***b* are disposed at the same position as the communication device **40** in the tire radial direction B. Furthermore, in this specification, the light emission area **61** of the first label area **50***a* and the light emission area **81** of the second label area **50***b* illustrated in FIGS. **1** to **3**, **4**A, **4**D, **5**, **6**, **7**, and **8**A to **8**E are disposed at the same position as the communication device **40** in the tire circumferential direction C. In this specification, the light emission area **91** of the third label area **50***c* illustrated in FIGS. **6**, **7**, and **8**B is disposed at the same position as the communication device **40** in the tire circumferential direction C.

The tire according to the present disclosure is not limited to the specific configurations described in the above embodiments and variations, and various variations, modifications, and combinations are possible without departing from the scope of the claims.

## INDUSTRIAL APPLICABILITY

The present disclosure relates to a tire.

## REFERENCE SIGNS LIST

**1**, **100**, **200** tire
**2** bead core
**3** bead filler
**4** carcass
**4***a* carcass ply
**4***a***1** body part
**4***a***2** folded part
**5** belt
**5***a*-**5***d* first to fourth belts
**7** tread rubber
**7***a* center land area
**7***a***1** inner circumferential groove
**7***b* intermediate land area
**7***b***1** outer circumferential groove
**7***c* shoulder land area
**8** side rubber
**9** inner liner
**10** tire body
**10***a* tread portion
**10***b* side portion
**10***b***1** first side portion
**10***b***2** second side portion
**11** sidewall portion
**12** bead portion
**13***a* tread outer surface (example of outer surface of tire body)
**13***b* tread inner surface (example of inner surface of tire body)
**14***a* side outer surface (example of outer surface of tire body)
**14***a***1** first side outer surface
**14***a***2** second side outer surface
**14***b* side inner surface (example of inner surface of tire body)
**14***b***1** first side inner surface
**14***b***2** second side inner surface
**15** tread end
**20***a*-**20***d* first to fourth belt layers
**40** communication device
**50** label area
**50***a* first label area
**50***a***1** vertex
**50***b* second label area
**50***c* third label area
**50***c***1** vertex
**61** light emission area of first label area
**62** non-light emission area of first label area
**68** ridge
**69** base protrusion portion
**70** convex portion
**71** convex portion
**81** light emission area of second label area
**91** light emission area of third label area
A tire width direction
B tire radial direction
B**1** outer side in tire radial direction
B**2** inner side in tire radial direction
C tire circumferential direction
D tire wall thickness direction
O central axis of tire
CL tire equatorial plane
L**1**, L**3**, L**4** virtual circle
L**2**, L**5**-L**10** virtual straight line

The invention claimed is:

**1**. A tire comprising:
a tire body including a side portion; and
a communication device embedded within the side portion of the tire body or mounted on an inner surface of the side portion of the tire body,
wherein
the tire body comprises a label area located on an outer surface of the side portion, the label area indicating a position of the communication device,
the tire body includes a cylindrical base protrusion portion which protrudes on the outer surface of the side portion, and an annular convex portion which protrudes from an outer edge of a top surface of the base protrusion portion, and
the label area comprises a light emission area which is provided on the top surface of the annular convex portion and a non-light emission area which is provided on the top surface of the base protrusion portion inside the annular convex portion.

**2**. The tire according to claim **1**, wherein the light emission area of the label area has a color or shape that can be distinguished from surroundings.

**3**. The tire according to claim **2**, wherein
the tire body comprises:
a tread portion; and
first and second side portions extending from both ends of the tread portion in a tire width direction toward an inner side in a tire radial direction,
wherein
the communication device is embedded within one of the tread portion, the first side portion, and the second side portion, or is attached to an inner surface of one of the tread portion, the first side portion, and the second side portion, and

the light emission area of the label area is disposed at a same position as the communication device in at least any one direction of the tire width direction, the tire radial direction, or a tire circumferential direction.

**4**. The tire according to claim **1**, wherein

the tire body comprises:

a tread portion; and

first and second side portions extending from both ends of the tread portion in a tire width direction toward an inner side in a tire radial direction,

wherein

the communication device is embedded within one of the tread portion, the first side portion, and the second side portion, or is attached to an inner surface of one of the tread portion, the first side portion, and the second side portion, and

the light emission area of the label area is disposed at a same position as the communication device in at least any one direction of the tire width direction, the tire radial direction, or a tire circumferential direction.

**5**. The tire according to claim **1**, wherein the communication device is disposed in such a position as not to overlap with the light emission area in a tire wall thickness direction, and is disposed in such a position as to overlap with the non-light emission area in the tire wall thickness direction.

**6**. The tire according to claim **1**, wherein

the non-light emission area is provided on a fine uneven surface,

the fine uneven surface comprises a base surface and a plurality of ridges arranged in parallel on the base surface, the ridges having a height of 0.1 to 1.0 mm, and

a distance between tops of two adjacent ridges of the plurality of ridges is 0.3 to 1.5 mm.

**7**. The tire according to claim **6**, wherein

the light emission area is formed by a resin phosphorescent material configuring a part of the outer surface of the tire body, the resin phosphorescent material carrying a phosphorescent material in void spaces of a foam resin,

the non-light emission area has a different color from the light emission area, and

the non-light emission area is formed of a resin coloring body that constitutes a part of the outer surface of the tire body, the resin coloring body carrying pigment in void spaces of a foam resin.

**8**. The tire according to claim **1**, wherein

the light emission area is formed by a resin phosphorescent material configuring a part of the outer surface of the tire body, the resin phosphorescent material carrying a phosphorescent material in void spaces of a foam resin,

the non-light emission area has a different color from the light emission area, and

the non-light emission area is formed of a resin coloring body that constitutes a part of the outer surface of the tire body, the resin coloring body carrying pigment in void spaces of a foam resin.

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